

Green subsidies with demand distortions

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Abstract

Demand distortions such as credit constraints and behavioral biases are pervasive in lower-income contexts. A model of clean technology adoption shows that distortions can lower abatement but increase the welfare gains from green subsidies, through two channels: shifting the marginal adopter toward higher private and social benefits, and increasing demand elasticity. We test this by cross-randomizing fixed-cost subsidies, marginal-cost subsidies, and loan access for a US\$84 induction stove among 2,134 charcoal users in Kenya. Marginal-cost subsidies of 75% have a precise zero effect on adoption and usage. Credit constraints, however, increase the observable welfare gain per fixed-cost subsidy dollar by between 30%–73% and lower the subsidy cost per tCO₂e from US\$20 to US\$12. This happens through the two hypothesized channels: increasing the marginal adopter’s positive externality by 19% and fuel savings by 30%, and lowering the subsidy cost per marginal adoption by 29%. We estimate the model to simulate an undistorted counterfactual: the subsidy cost per tCO₂e rises to US\$160. Climate financing could be most effective in contexts with larger demand distortions.

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1 Introduction

Many climate change mitigation technologies generate social benefits only if an initial investment is followed by a stream of follow-through decisions: an electric vehicle abates emissions only when driven in place of a gasoline car. When follow-through generates private benefits, as with most energy-saving technologies, the market selects well: the agents who generate the largest social benefits also have the highest willingness-to-pay (WTP) and adopt first.¹ However, this same force creates an adverse selection problem for subsidies. Only agents with low private benefits would choose not to adopt, so adopters who are marginal to the subsidy have lower usage and generate less social benefit. Much of the green subsidy expenditure accrues to inframarginal agents, who would have adopted anyway. These dynamics have dampened the social benefits of a wide range of programs subsidizing upfront investment.² A marginal-cost subsidy aims to avoid these problems: by aligning private incentives with social benefits, it draws in adopters in proportion to the abatement they generate and encourages follow-through on the intensive margin (Pigou, 1920).

This logic assumes that WTP reflects private benefit. However, in many settings demand distortions drive WTP below the private benefit (Atkin et al., 2025).³ Behavioral biases such as present bias, loss aversion, and inattention cause agents to undervalue benefits (Laibson, 1997; DellaVigna, 2009; Duflo et al., 2011). Capital market failures prevent agents from accessing future payment streams to fund present-day investments (de Mel et al., 2008; Banerjee and Duflo, 2014). Transaction costs and savings constraints can distort subsidy targeting in household energy purchasing (Abubakari et al., 2026). As a result, even agents with large private benefits from follow-through may not make the initial investment, leaving them marginal to a subsidy. Relative to an undistorted market, distortions thus make green subsidies more effective through two channels: each dollar can induce more adoptions, and each induced adoption can generate larger social benefits. At the same time, these distortions can generate real economic cost, undermining the welfare gains subsidies generate.

To formalize this logic, we develop a simple model of how demand distortions affect the optimal use of marginal-cost subsidies (which operate on the follow-through margin) and fixed-cost subsidies (which operate on the investment margin) for incentivizing clean technology adoption. The model yields two results: subsidies generate larger welfare gains in contexts with larger demand distortions, and fixed-cost subsidies can dominate marginal-cost subsidies in such contexts. We characterize conditions under which these results hold even if the distortion adds real economic costs.

We evaluate these dynamics empirically in the context of induction cookstove adoption among 2,134 charcoal-cooking households in Nakuru, Kenya. This setting exhibits severe capital market

¹When follow-through is privately costly, as with forest conservation or sustainable agricultural practices, subsidizing the investment attracts the agents least willing to follow through, and contracting directly on follow-through requires costly information and enforcement capacity and may introduce moral hazard (Oliva et al., 2020; Jack et al., 2025; Aspelund and Russo, 2026; Chen et al., 2025).

²See for example Boomhower and Davis (2014), Rapson and Muehlegger (2023), and Hahn et al. (2026).

³Some papers refer to the observable object as ability-to-pay and the unobservable, welfare-relevant object as WTP (e.g. Banerjee, 1997). Other papers, including ours, use WTP to denote observed demand (e.g. Cohen and Dupas, 2010; Tarozi et al., 2014; Burlig et al., 2026).

distortions: even with sophisticated credit technologies, buyers paying in installments are normally charged an annualized percentage rate (APR) of 224%. This allows us to estimate the impact and optimal design of green subsidies by (i) varying fixed- and marginal-cost subsidies; (ii) manipulating demand distortions; and (iii) using the resulting random variation to causally estimate the private and social benefits of a green technology.

We cross-randomize fixed-cost subsidies of either 10% or 75% off the US\$84 price and marginal-cost subsidies that lower the price of electricity used to operate the stove by 0%, 25%, or 75% for six months (with no change in the cost of electricity for other uses). We randomize access to an interest-free, four-month loan designed to reduce the wedge between WTP and private benefit. We elicit WTP using an incentive-compatible Becker et al. (1964) mechanism. 29% of study participants bought the stove bundle (which included three pieces of compatible cookware), of whom 67% bought it with a loan, with repayment averaging 86%. We convert fuel expenditures to tons of carbon dioxide-equivalent (tCO_2e) using market prices and local deforestation rates.

We first quantify the induction stove’s private and social impacts. High-frequency temperature sensors record a reduction in daily charcoal cooking of 25 minutes, and self-reported monthly charcoal spending decreases by US\$10. Self-reported monthly electricity spending increases by US\$2.3, which aligns with the induction stove’s independent measurements (15 kWh per month at US\$0.166 per kWh). Aggregate changes in biomass, electricity, and LPG usage save US\$8.9 per month and abate 2.5 tCO_2e per year. Historical usage data from the manufacturer suggest a 2.2-year lifespan.

We find that marginal-cost subsidies have no impact. Despite providing detailed information and drawing attention to cooking costs before eliciting WTP, each US\$1 in electricity subsidy has a precisely estimated US\$0.0 impact on WTP (95% CI: $[-\text{US}\$0.2, \text{US}\$0.2]$), and we can rule out that a 1% usage price decrease increases electric stove usage by more than 0.12%. These results hold with and without access to a loan.

Each fixed-cost subsidy dollar generates US\$3.4 in fuel savings and US\$10 in environmental benefit when valuing CO_2e at a social cost of carbon (SCC) of US\$120 per ton (EPA, 2023). The subsidy cost per tCO_2e , US\$12, is roughly one-tenth of the SCC. The economic cost of abatement is *negative*: fuel savings exceed the stove’s production cost. We estimate that each subsidy dollar generates an observable welfare gain of US\$10–US\$12 depending on how much of the distortion generates resource costs.

The paper’s main finding is that demand distortions substantially increase the welfare gains that subsidies generate. Average WTP in the control group is only 9.3% of the net present value of future discounted fuel savings: when given access to an interest-free loan this increases to 17%. We find that demand distortions in this context increase the observable welfare benefit of fixed-cost subsidies substantially, from US\$7–US\$8 up to US\$10–US\$12 per dollar of government spending. They furthermore decrease the subsidy cost per tCO_2e from US\$20 to US\$12.

Distortions operate through both channels identified in the theory. First, marginal adopters facing larger credit distortions generate 30% more in private fuel savings (US\$175 versus US\$227) and abate 19% more CO_2e (4.8 tCO_2e versus 5.7 tCO_2e). Second, credit constraints shift demand

elasticity from -0.9 to -2.3 , reducing inframarginal expenditure from 38% to 6.5% of total subsidy spending. This reduces the subsidy expenditure per marginal sale by 29%, from US\$96 to US\$67.

The experiment only partially relaxes distortions: even with loan access, WTP remains well below discounted private benefits. How efficient would fixed-cost subsidies be in a setting without distortions, where WTP equals the private benefit? To answer this, we estimate our model using two-step generalized method of moments (GMM) and simulate counterfactuals across a range of distortions. The model is identified using the randomized variation from the experiment. Simulations indicate that fully eliminating distortions up to the efficient WTP would raise the subsidy cost per tCO_{2e} from US\$12 to US\$160 (14 times higher).

Annual carbon mitigation financing exceeds US\$1 trillion (Buchner, 2024). Efficient abatement requires allocating each marginal dollar towards the lowest cost abatement technology. Our results suggest that many of the lowest cost portions of the global abatement supply curve could be found in contexts with large market distortions. For example, the U.S. has a sophisticated financial market for consumer goods, with 80% of new cars sold with financing (Experian, 2023). Through the mechanisms identified in this paper, this absence of credit constraints could partly explain why U.S. electric vehicle subsidies cost over US\$1,200 in government expenditure per tCO_{2e} abated (Hahn et al., 2026)—100 times the US\$12 per ton in our setting. Even holding the underlying technology fixed, our results indicate that climate mitigation expenditure could generate substantially larger welfare gains in contexts with more distorted markets.

More than 2 billion people lack modern cooking technologies (IEA, 2024). With billions gaining access to electricity in recent decades, widespread electric cooking has become an achievable policy goal (World Bank, 2024). Despite the importance of this problem, causal evidence is scarce. Using experimental variation, Desbureaux et al. (2025) and Mahadevan et al. (2025) find that fixed-cost subsidies increase adoption of electric cooking technologies in the DRC and in Nepal, respectively, which increases electricity usage and generates important social benefits. Gould et al. (2023) find that the nationwide transition from gas towards electric cooking reduced Ecuador’s CO_{2e} emissions. Other descriptive research and laboratory experiments have shed light on perceptions, usage patterns, energy efficiency, and potential for scale of both induction stoves and pressure cookers (Aemro et al., 2021; Leary et al., 2021; Saha et al., 2021; Clements et al., 2020; Alem et al., 2014; Banerjee et al., 2016; Rubinstein et al., 2022).

While Pigouvian taxes are theoretically first-best (Pigou, 1920), this paper primarily focuses on green subsidies for several reasons. Carbon taxes are often politically intractable: in Canada, a well-designed carbon tax was recently repealed due to political pressures (Department of Finance Canada, 2025). Energy expenditures tend to be inelastic, and low-income households spend a disproportionate share of their incomes on energy costs: in June 2026, Kenya froze electricity tariffs amid “public discontent over fuel prices” (Daily Nation, 2026). Taxation may also be logistically infeasible given the widespread informality of firewood and charcoal production. In addition, many environmental principals (e.g. multilateral donors, environmental agencies, the UNFCCC, carbon offset markets, philanthropists, and investors) lack the legal authority to levy a tax. Their focus

is subsidizing climate change mitigation. Doing so efficiently requires identifying technologies and contexts with the lowest marginal abatement costs, at the bottom of the abatement supply curve, and then moving up over time (Gillingham and Stock, 2018; Kolstad, 2011).

Related literature This paper contributes to a new but rapidly growing literature documenting how demand distortions, or wedges, affect market outcomes, including for example Atkin et al. (2025), Bergquist et al. (2026), Ghatak and Mookherjee (2025), and Indarte et al. (2026). This new framework builds on a large literature in development economics documenting that pervasive market failures dampen WTP to below private welfare, and that credit constraints in particular depress adoption of privately beneficial technologies.⁴

We evaluate demand distortions in the context of selection and mechanism design for optimal subsidy design. This literature evaluates how the relationship between agents’ private willingness to participate and the social value of their actions affects the welfare consequences of subsidy programs (Aspelund and Russo, 2026; Chen et al., 2025; Ito et al., 2023). We build on prior research showing that demand distortions can change the efficiency properties of subsidy instruments (Abubakari et al., 2026). Many existing subsidy efficacy results (e.g. Allcott and Taubinsky, 2015; Gillingham and Palmer, 2014; Allcott et al., 2014) study allocation distortions, which only affect payoffs through adoption and usage decisions. The distortions prevalent in developing countries, however—like credit constraints and transaction costs—often incur real economic costs.

Third, we contribute to a longstanding literature on optimal environmental policy instruments and second-best environmental policies in the presence of two or more market failures (Jacobsen et al., 2020; Knittel and Sandler, 2018; Holland et al., 2016; Fowlie et al., 2016; Allcott and Taubinsky, 2015). While some papers have shown the benefits of fixed-cost subsidies over marginal-cost subsidies in some contexts (e.g. Carlton and Loury, 1980; De Groote and Verboven, 2019), we show that demand distortions such as behavioral biases and credit constraints can *increase* the efficiency of green subsidies.

Finally, a large empirical energy economics literature, primarily focused on high-income countries, studies intensive and extensive margin demand elasticities (Ito and Zhang, 2025; Ito, 2014; Kahn and Wolak, 2013; Ida et al., 2026; Hahn and Metcalfe, 2021; Houde and Aldy, 2017; Boomhower and Davis, 2014; Borenstein and Davis, 2025). We estimate both margins in a very different context and find a substantially more elastic extensive margin response to fixed cost subsidies, but we detect no extensive margin or intensive margin response to marginal-cost subsidies. This advances the policy debate around electricity tariffs designed to spur decarbonization in low- and middle-income countries. Our research design circumvents a technological constraint of subsidies inadvertently incentivizing the use of electricity for other purposes (Borenstein, 2025; Campbell, 2025).

⁴Cohen and Dupas (2010), Tarozzi et al. (2014), Burlig et al. (2026), Banerjee and Duflo (2014), de Mel et al. (2008), Fafchamps et al. (2014), McKenzie and Woodruff (2008), Duflo et al. (2008), Gertler et al. (2024), Adhvaryu et al. (2020), and Berkouwer and Dean (2022).

Table 1: Examples of distortions that lower willingness to pay

Distortion	$D(v)$	$\rho(P)$	Examples
<i>Distortions that may consume resources</i>			
Credit markup	$v/(1+r)$	$r \cdot P$	<i>This paper</i>
Hassle, transaction costs	$v - c$	c	Abubakari et al. (2026)
<i>Pure decision distortions ($\rho = 0$)</i>			
Present bias, inattention	$(1 - \theta)v$	–	Duflo et al. (2011); Allcott and Wozny (2014)
Additive bias	$v - b$	–	Allcott and Taubinsky (2015)
Ironing	$\mathbb{E}[w \mid w \leq v]$	–	Ito (2014); Rees-Jones and Taubinsky (2020)
Convex misperception	$v^2/\bar{v}$	–	Bushong et al. (2021)

Notes: Examples of increasing mappings that lie weakly below the identity and are strictly below it over the relevant interior. The parameters satisfy $r, c, b, v > 0$ and $\theta \in (0, 1)$. w denotes the value of an inframarginal unit; $\bar{v}$ denotes the maximum willingness to pay in the support. Decision distortions have $\rho = 0$.

2 Theory

We evaluate a green technology that requires purchase to enable follow-through, where follow-through generates a positive externality. A principal can offer either a usage subsidy, which lowers the marginal cost of following through, or a purchase subsidy, which lowers the cost of buying the technology.

How does a demand distortion change the return to a purchase subsidy? And, when can that subsidy outperform a usage subsidy? We first isolate two behavioral channels through which distortions can affect marginal and average returns in a generalized framework. We then apply a specialized version of the model to our study setting and compare the first-dollar returns.

2.1 A generalized framework

Consider a mass of agents indexed by their undistorted willingness to pay v for adopting a green technology, with continuous distribution F , density f , and survival function $Q(v) \equiv 1 - F(v)$. Let S denote the external benefit generated by an adoption, and define the conditional external-benefit profile

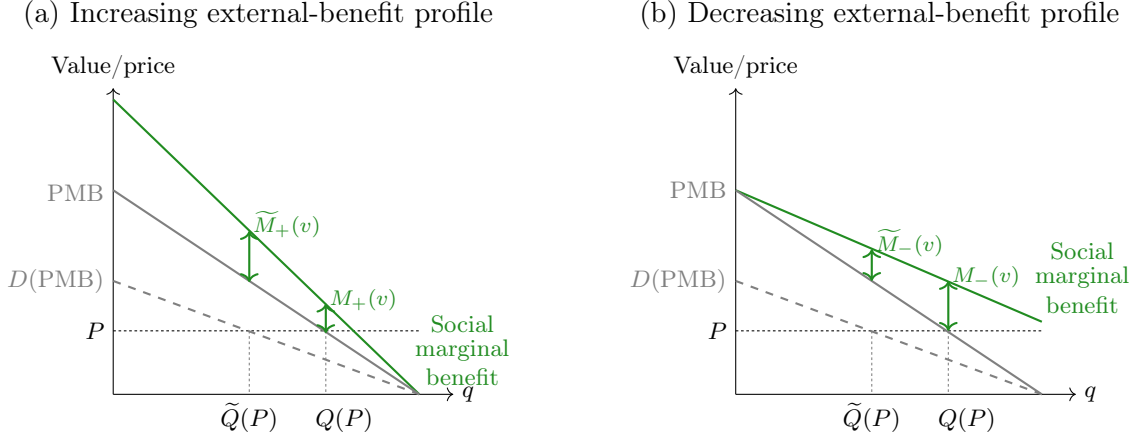
$$M(v) \equiv \mathbb{E}[S \mid v]. \quad (1)$$

A demand distortion maps frictionless willingness to pay v into the decision-relevant WTP $D(v)$, where $D(v) \leq v$.⁵ Let p be the fixed market price, let s_1 be a fully passed-through purchase subsidy, and let $P = p - s_1$ denote the net purchase price faced by the buyer. An agent adopts when $D(v) \geq P$. Write $\tilde{Q}(P) = \mathbb{P}(D(v) \geq P)$ for adoption under the distortion.

In some cases, such as inattention, the distortion affects decision-making but does not affect payoffs conditional on the actions taken. In other cases, the distortion also consumes real resources. Let $\rho(P)$ denote this resource cost per adopter (for a decision distortion write $\rho(P) = 0$).

⁵We assume that D is continuously differentiable with $D'(v) > 0$ over the relevant range and that all cutoffs are interior, with positive density and positive remaining demand.

Figure 1: How the external-benefit profile determines the role of selection



Notes: Social marginal benefit (SMB) equals private marginal benefit (PMB) plus the conditional external benefit. The labels $M(v)$ and $\tilde{M}(v)$ show this gap at the undistorted and distorted cutoffs, respectively. The distortion increases the gap when $M(v)$ is increasing ($\tilde{M}_+ > M_+$) and decreases it when $M(v)$ is decreasing ($\tilde{M}_- < M_-$).

Table 1 gives examples of distortions that fit this reduced-form representation. Although their underlying mechanisms differ, each can be summarized for the adoption decision by a mapping from undistorted willingness to pay v to distorted willingness to pay $D(v)$.

2.1.1 Channel 1: selection

For any given price P , the undistorted and distorted marginal adopters, $\underline{v}$ and $\tilde{v}$, satisfy

$$\underline{v} = P = D(\tilde{v}) \leq \tilde{v}. \quad (2)$$

The distortion weakly reduces adoption and moves the margin toward types with higher private values. Whether this raises or lowers the marginal externality depends on the slope of $M(v)$. If $M(v)$ is increasing over the relevant range, then the distortion increases the marginal adopter's externality ($M(\tilde{v}) \geq M(\underline{v})$). This reverses if $M(v)$ is decreasing. Figure 1 illustrates this.

2.1.2 Channel 2: demand elasticity

The demand elasticity determines how much of each purchase subsidy dollar causes additional adoptions. A marginal increase ds_1 in the purchase subsidy induces $-Q'(P) ds_1$ new adoptions and raises expenditure on existing adopters by $Q(P) ds_1$. Let κ denote the number of inframarginal adopters per induced adoption and let $|\mathcal{E}(P)| \equiv -PQ'(P)/Q(P)$ denote the price elasticity:

$$\frac{\tilde{\kappa}(P)}{\kappa(P)} = \frac{\tilde{Q}(P) / \tilde{Q}'(P)}{Q(P) / Q'(P)} = \frac{|\mathcal{E}(P)|}{|\tilde{\mathcal{E}}(P)|}. \quad (3)$$

Thus, if $|\mathcal{E}(P)| < |\tilde{\mathcal{E}}(P)|$ then $\kappa(P) < \tilde{\kappa}(P)$: a distortion that raises elasticity lowers inframarginal subsidy expenditure per induced adoption by the inverse proportion.

2.1.3 Return to a purchase subsidy

Let $\mathcal{V}(v)$ denote the contribution of adoption by type v to the principal's objective, excluding any resource cost. It may include only the externality or, for a social-welfare objective, private benefits net of the technology's real economic cost. Let W denote the aggregate objective, and let C denote subsidy expenditure. A marginal increase in s_1 induces agents at $\tilde{v} = D^{-1}(P)$. The principal spends $s_1 + \tilde{\kappa}(P)$ per induced adoption: s_1 on the new adopter and $\tilde{\kappa}(p - s_1)$ on existing adopters.

Aggregate resource use $R(s_1)$ includes the resource cost $\rho(P)$ of inframarginal and induced adoptions. A subsidy induces resource cost among induced adopters it but lowers the resource cost among inframarginal adopters:

$$R(s_1) \equiv \rho(p - s_1)\tilde{Q}(p - s_1), \quad \dot{R}(s_1) = \frac{dR/ds_1}{d\tilde{Q}/ds_1} = \underbrace{\rho(p - s_1)}_{\text{induced adopters' resource cost}} - \underbrace{\rho'(p - s_1)\tilde{\kappa}(p - s_1)}_{\text{inframarginal adopters resource savings}} \quad (4)$$

Taken together, the principal's welfare return to an additional purchase subsidy dollar is

$$\frac{dW/ds_1}{dC/ds_1} = \frac{\mathcal{V}(\tilde{v}) - \dot{R}(s_1)}{s_1 + \tilde{\kappa}(P)}. \quad (5)$$

For a finite subsidy relative to no subsidy, let ΔQ denote the number of adoptions induced by the subsidy, ζ the number of subsidy recipients per induced adoption, $\bar{\mathcal{V}}$ the average value of an induced adoption, and ΔR the change in aggregate resource use. Average welfare generated per subsidy dollar is

$$\frac{\Delta W}{\Delta C} = \frac{\bar{\mathcal{V}} - \Delta R/\Delta Q}{s_1\zeta}. \quad (6)$$

The marginal and average cases have the same structure: selection determines the value of induced adopters, elasticity determines expenditure per induced adoption, and resource costs enter separately.

2.1.4 Return to a usage subsidy

We now allow the principal to subsidize usage at a per-unit rate $s_2 \geq 0$ and evaluate both instruments at the first dollar, $s_1 = s_2 = 0$. Let $x(v)$ denote usage conditional on adoption in the absence of a subsidy, with $x'(v) > 0$. A marginal usage subsidy raises type v 's willingness to pay for adoption by their usage $x(v)$,⁶ and it is paid on the usage of every adopter.

The usage subsidy does not change the selection result: the distortion still weakly raises the undistorted private value of the marginal adopter. The two instruments differ in how far each dollar moves the adoption margin and in what it is paid on. Each dollar of usage subsidy raises the

⁶This is the envelope-theorem result when usage is chosen optimally. We assume that any attention distortion is homogeneous between future savings due to lower fuel costs and the subsidy. If agents are less attentive to usage subsidies than regular savings, they may not accurately perceive the usage subsidy's incentive as being less than $x(v)$.

marginal adopter's WTP by $x(\tilde{v})$ and is paid on average usage $\bar{x}$ per adopter.⁷ Induced adoptions per subsidy dollar are therefore

$$\left. \frac{dQ/ds_1}{dC/ds_1} \right|_{s_1=s_2=0} = \frac{1}{D'(\tilde{v})} \frac{f(\tilde{v})}{Q(\tilde{v})}, \quad \left. \frac{dQ/ds_2}{dC/ds_2} \right|_{s_1=s_2=0} = \frac{x(\tilde{v})}{\bar{x}} \frac{f(\tilde{v})}{Q(\tilde{v})}. \quad (7)$$

The marginal adopter uses weakly less than the average adopter, so $x(\tilde{v})/\bar{x} \leq 1$, while $1/D'(\tilde{v}) \geq 1$ whenever $D'(\tilde{v}) \leq 1$. As a result, the purchase subsidy induces weakly more adoptions per dollar.

The usage subsidy's advantage, however, is that it can change usage among existing adopters. Let $\mathcal{V}_2(v, 0)$ denote the marginal effect of s_2 on the principal's value from type v at $s_2 = 0$, and define

$$\mathcal{I}_2 \equiv \frac{\int_{\tilde{v}}^{\infty} \mathcal{V}_2(v, 0) dF(v)}{\tilde{Q}(p) \bar{x}}, \quad (8)$$

the value generated per subsidy dollar by this intensive-margin effect, excluding resource costs. This is the object the credit comparison takes as given.

Let $R(s_1, s_2)$ denote aggregate resource use, with $R(s_1, 0) = R(s_1)$, and let R_1 and R_2 denote its partial derivatives at $s_1 = s_2 = 0$. A marginal increase in s_2 induces $f(\tilde{v}) x(\tilde{v})$ adoptions, changes the principal's value from existing adopters by $\int_{\tilde{v}}^{\infty} \mathcal{V}_2(v, 0) dF(v)$, and costs $\tilde{Q}(p) \bar{x}$. The first-dollar return to the usage subsidy is therefore

$$\left. \frac{dW/ds_2}{dC/ds_2} \right|_{s_1=s_2=0} = \underbrace{\frac{x(\tilde{v})}{\bar{x}} \frac{f(\tilde{v})}{Q(\tilde{v})} \mathcal{V}(\tilde{v})}_{\text{extensive margin}} + \underbrace{\mathcal{I}_2}_{\text{intensive margin}} - \frac{R_2}{\tilde{Q}(p) \bar{x}}. \quad (9)$$

Evaluating the return of the previous section at $s_1 = 0$ gives the analogous expression

$$\left. \frac{dW/ds_1}{dC/ds_1} \right|_{s_1=s_2=0} = \frac{1}{D'(\tilde{v})} \frac{f(\tilde{v})}{Q(\tilde{v})} \mathcal{V}(\tilde{v}) - \frac{R_1}{\tilde{Q}(p)}. \quad (10)$$

Both returns have the same structure: induced adoptions per dollar, times the value of the marginal adopter, net of the change in resource use per dollar. Selection enters both through $\mathcal{V}(\tilde{v})$, since the two instruments induce the same marginal type. They differ in the extensive-margin weight, $x(\tilde{v})/\bar{x}$ versus $1/D'(\tilde{v})$, and in the intensive-margin term, which only the usage subsidy has. Comparing the two, the usage subsidy earns a higher first-dollar return if and only if

$$\mathcal{I}_2 - \frac{R_2}{\tilde{Q}(p) \bar{x}} + \frac{R_1}{\tilde{Q}(p)} > \left[\frac{1}{D'(\tilde{v})} - \frac{x(\tilde{v})}{\bar{x}} \right] \frac{f(\tilde{v})}{Q(\tilde{v})} \mathcal{V}(\tilde{v}). \quad (11)$$

⁷ D' does not appear in the expression because a distortion that dampens the usage subsidy's pass-through into decision-relevant willingness to pay also compresses agents along that scale. The smaller movement in decision-relevant WTP therefore crosses proportionally more agents near the adoption cutoff, so the two effects cancel.

$$\begin{array}{ccccccc}
\underbrace{\mathcal{I}_2}_{\text{intensive-margin}} & + & \underbrace{\frac{R_1}{\tilde{Q}(p)}}_{\text{avoided resource}} & - & \underbrace{\frac{R_2}{\tilde{Q}(p)\bar{x}}}_{\text{resource cost}} & > & \left[\underbrace{\frac{1}{D'(\tilde{v})}}_{\text{marginal WTP}} - \underbrace{\frac{x(\tilde{v})}{\bar{x}}}_{\text{marginal WTP}} \right] \underbrace{\frac{f(\tilde{v})}{Q(\tilde{v})}}_{\text{density of}} \underbrace{\nu(\tilde{v})}_{\text{value of the}} \\
\text{value per usage-} & & \text{cost per} & & \text{per usage-} & & \text{increase per} & \text{increase per} & \text{marginal} & \text{value of the} \\
\text{subsidy dollar} & & \text{purchase subsidy} & & \text{subsidy dollar} & & \text{purchase subsidy} & \text{usage subsidy} & \text{adopters} & \text{marginal} \\
& & \text{dollar} & & & & \text{dollar} & & & \text{adopter}
\end{array} \tag{12}$$

Each term has an economic interpretation. The left hand side is the intensive margin benefit of the usage subsidy plus the difference in induced resource costs between the purchase and usage subsidies. The right hand side is the difference in the value of adoptions induced by the purchase subsidy and the usage subsidy. The purchase subsidy outperforms the usage subsidy whenever its extensive margin advantage outweighs the usage subsidy's intensive margin effects net of the difference in resource costs.

2.2 Credit constraints and energy-efficiency adoption

We now specialize the framework to the credit constraint studied in the application, leveraging three defining features of the context. First, we characterize the credit market failure as a resource cost $r \geq 0$ that raises the cost of financing that purchase to $(1+r)P$ and generates a resource cost rP per purchase. The agent adopts when

$$v_i \geq (1+r)P, \quad \text{or equivalently} \quad D(v_i) = \frac{v_i}{1+r} \geq P. \tag{13}$$

Aggregate distorted demand is then given by $\tilde{Q}(P) = Q((1+r)P)$.

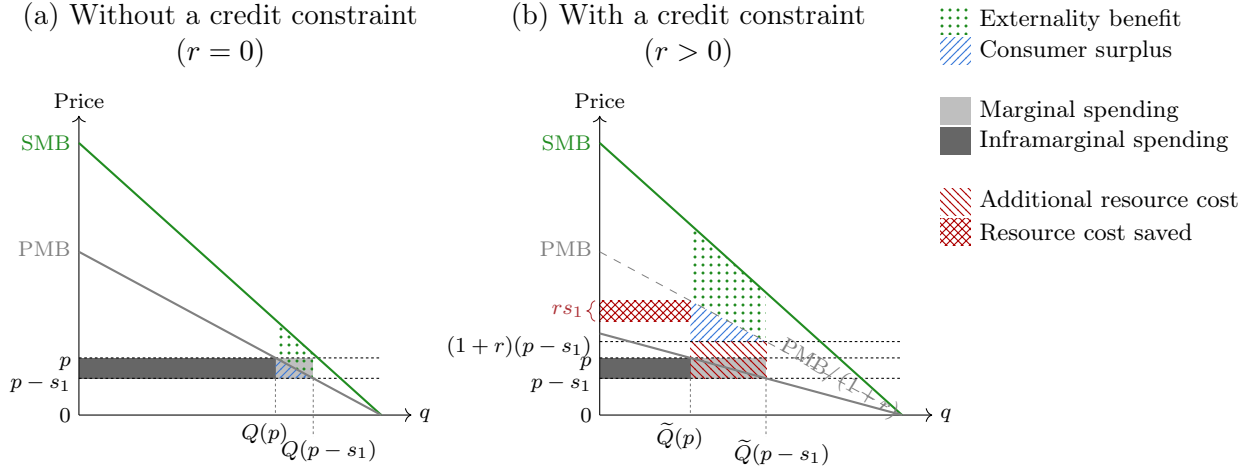
Second, we assume that fuel savings and emissions reductions both scale with increased displacement of charcoal. While these vary with emissions factors and market prices, aggregate emissions and fuel spending have an R^2 of 0.6 in our context.

Third, we model total cooking as being perfectly inelastic so that the only usage decision is which technology to use rather than whether to cook at all. In our data, we detect no change in the number of meals cooked per day (95% CI of [-0.06, 0.18] on a control mean of 2.8) after adoption.

Figure 2 summarizes how a credit constraint changes the effects of a purchase subsidy, using a pared-down version of the model to build graphical intuition. Let ω_i denote agent i 's clean-technology usage after adoption (in the absence of a usage subsidy, which we introduce later). Each unit displaces the incumbent technology by an amount that saves π in fuel costs and generates an external benefit ϕ . Thus agent i 's private value is $v_i = \pi\omega_i$, and their total external benefit is $S_i = \phi\omega_i$.

The distortion increases subsidy efficacy through two channel. Through the selection channel, the demand distortion causes any given purchase subsidy s_1 to generates a larger marginal positive externality. Through the elasticity channel, it causes an increase in the number of adoptions to requires a smaller increase in subsidy spending, as long as $f(v)/Q(v)$ is increasing. In addition,

Figure 2: Private value, social value, resource cost, and expenditure under a purchase subsidy



Notes: Private marginal benefit (PMB) and Social marginal benefit (SMB). The dotted green area is the externality generated by induced adopters, the striped blue area is their consumer surplus, and the gray areas are subsidy expenditure. In panel (b), the striped red area is the resource cost incurred by induced adopters. The crosshatched area is the resource cost saved when the subsidy reduces the amount financed for inframarginal adopters. The gray areas measure government expenditure; they are not themselves resource costs.

induced adopters incur an additional resource cost $r(p-s_1)$ but inframarginal adopters' resource cost goes down by rs_1 .

2.2.1 Welfare effects of a purchase subsidy

Let $\Delta Q \equiv \tilde{Q}(p-s_1) - \tilde{Q}(p)$ denote the number of adoptions induced by the subsidy, and define

$$\zeta(r) \equiv \frac{\tilde{Q}(p-s_1) - \tilde{Q}(p)}{\Delta Q} \quad (14)$$

as the number of subsidy recipients per induced adoption. The ΔQ induced adoptions generate total resource cost $r(p-s_1)\Delta Q$ (the striped red area in panel (b)), while the subsidy saves total resource cost $rs_1\tilde{Q}(p)$ among inframarginal adopters (the crosshatched area).⁸

Define total subsidy expenditure $\Delta C = s_1\tilde{Q}(p-s_1)$. There are $\zeta(r) - 1$ inframarginal adopters per induced adoption. The average welfare gain per subsidy dollar is then given by⁹

$$\mathcal{W}(r) \equiv \frac{\Delta W}{\Delta C} = \frac{\bar{\mathcal{V}}(r) - r(p-s_1) + rs_1(\zeta(r) - 1)}{s_1\zeta(r)} = \frac{\bar{\mathcal{V}}(r) - rp}{s_1\zeta(r)} + r. \quad (15)$$

The return is positive *iff* the additional resource cost for induced adopters does not outweigh the other benefits: $\bar{\mathcal{V}}(r) + rs_1\zeta(r) > rp$.

⁸This assumes buyers finance the full net purchase price. Subsidy payments are treated as transfers, as above. If only part of the distortion is a resource cost, the same accounting applies to that part.

⁹We assume that the relevant objects are differentiable and that the cutoffs remain interior in a neighborhood of r ; primes denote derivatives with respect to r .

The marginal effect of the financing distortion on welfare per subsidy dollar is:

$$\mathcal{W}'(r) = \frac{1}{s_1 \zeta(r)} \left[\underbrace{\bar{\mathcal{V}}'(r)}_{\text{selection benefit}} - \underbrace{[\bar{\mathcal{V}}(r) - rp] \frac{\zeta'(r)}{\zeta(r)}}_{\text{elasticity effect}} + \underbrace{s_1(\zeta(r) - 1)}_{\text{resource savings on inframarginal recipients}} - \underbrace{(p - s_1)}_{\text{resource cost of the induced purchase}} \right] \quad (16)$$

When social value is increasing in undistorted WTP over the relevant range, a larger distortion raises $\bar{\mathcal{V}}$ by moving the induced group toward higher-value agents (the selection channel). When $f(v)/Q(v)$ is increasing it weakly lowers $\zeta(r)$ by increasing the share of recipients induced by the subsidy, which reduces expenditure per induced adoption (the elasticity channel).

The resource savings effect arises because a larger distortion makes each dollar price reduction more valuable. Working in the opposite direction, a larger distortion also raises the resources required for each induced purchase, producing the final term. Thus the return is locally increasing in r as long as the additional resource cost does not outweigh the other three mechanisms.¹⁰

2.2.2 When can a purchase subsidy outperform a usage subsidy?

Specializing the general first-dollar expressions to the credit distortion, $D(v) = v/(1+r)$, gives $1/D' = 1+r$. Moreover, because $v_i = \pi\omega_i$, baseline usage $x(v)$ is increasing in v , so $x(\tilde{v})/\bar{x} \leq 1$. It follows that

$$\left. \frac{dQ/ds_1}{dC/ds_1} \right|_{s_1=s_2=0} = (1+r) \frac{f(\tilde{v})}{Q(\tilde{v})} \geq \frac{x(\tilde{v})}{\bar{x}} \frac{f(\tilde{v})}{Q(\tilde{v})} = \left. \frac{dQ/ds_2}{dC/ds_2} \right|_{s_1=s_2=0}. \quad (17)$$

Intuitively, a usage subsidy gives the marginal adopter an incentive proportional to their usage but makes still larger payments to higher-usage inframarginal adopters. A purchase subsidy pays every adopter the same amount, and credit amplifies its effect on the adoption threshold. The purchase subsidy therefore induces weakly more adoptions per dollar.

Both instruments induce the same marginal type, but only the usage subsidy changes usage conditional on type, generating the intensive-margin return $\mathcal{I}_2$ among existing adopters. Credit introduces two additional resource effects. Each induced purchase incurs financing resource cost rp , while each purchase-subsidy dollar paid to existing adopters saves r in financing resources. The purchase subsidy therefore has the larger first-dollar welfare return if and only if

$$[\mathcal{V}(\tilde{v}) - rp] \left[(1+r) - \frac{x(\tilde{v})}{\bar{x}} \right] \frac{f(\tilde{v})}{Q(\tilde{v})} + r > \mathcal{I}_2. \quad (18)$$

The first term is the value of the purchase subsidy's additional induced adoptions, net of their financing resource cost; it favors the purchase subsidy when $\mathcal{V}(\tilde{v}) > rp$. The term r is the financing resource saving on existing purchases. The purchase subsidy outperforms when these extensive-margin and resource-saving gains exceed the usage subsidy's intensive-margin return. A sufficiently

¹⁰A comparison between two values of r requires this condition to hold over the intervening range.

large $\mathcal{I}_2$ reverses the ranking.

2.2.3 Model extensions

We now consider several model extensions that might affect these results.

Marginal cost of raising funds Under the same increasing-hazard-rate condition, introducing a positive marginal cost of raising funds would strengthen the advantage of subsidies in a distorted market, since the distortion reduces inframarginal spending (Appendix X).

Taxation With a tax available, first-best usage can be induced by setting the marginal relative-price wedge equal to the externality (Pigou, 1920). Let v_i^{FB} denote agent i 's private value of adoption evaluated after this correction of the usage margin. Under a pure allocation distortion, only the adoption margin remains distorted. The marginal first-best adopter satisfies

$$\underline{v}_{\text{FB}} = p.$$

To make that same type marginal under the distortion, set

$$D(\underline{v}_{\text{FB}}) = p - s_1,$$

which implies $s_1 = p - D(p)$. However, under a resource distortion, no purchase subsidy restores the first best because the distortion both affects the adoption decision and the resource costs reduce welfare directly. Eliminating the resource cost requires $s_1 = p$, making the technology free and generally inducing over-adoption; any smaller subsidy leaves some resource cost in place. The planner therefore chooses a second-best $s_1 > 0$, balancing the welfare loss from over-adoption at the margin against the resource cost saved on every adoption (appendix X). The model above could also accommodate a budget neutral policy if environmental tax revenues must be earmarked for an accompanying environmental subsidy.

Direct instruments If the principal can correct an allocation distortion directly, they should do so as long as the cost per induced adoption does not exceed inframarginal spending per marginal adoption (appendix X). Correcting a resource distortion directly is more advantageous since it also saves directly on resource cost.

Correcting the resource distortion Suppose the principal can spend resources to reduce r itself. A purchase subsidy saves $r\kappa$ in financing resources per induced adoption, whereas reducing r saves $(1+r)\kappa$: it lowers the adoption threshold and reduces the financing cost on every existing purchase. At equal policy cost per induced adoption, correcting r therefore dominates the purchase subsidy. Because correction generates greater resource savings, it can remain worthwhile even when its cost per induced adoption exceeds κ (Appendix X).

Subsidizing the resource distortion Alternatively, the principal can leave r unchanged but pay $\varsigma(p - s_1)$ of each buyer’s financing charge, so that the buyer faces the rate $r - \varsigma$ and adopts iff

$$v_i > (1 + r - \varsigma)(p - s_1).$$

This policy lowers the cost perceived by the buyer but does not eliminate the underlying financing resource cost. Per dollar of expenditure, a purchase subsidy moves the adoption threshold $1 + r$ times as far: each dollar of s_1 reduces both the purchase amount and the financing cost attached to that amount. The purchase subsidy therefore dominates subsidizing the financing charge on the extensive margin, increasingly so as r rises (Appendix X).

Heterogeneous distortions When the distortion is heterogeneous, different instruments no longer induce the same agents. The most distorted agents have the highest adoption thresholds, so an intervention targeted at them induces high-value adopters, whereas a uniform s_1 is offered across all distortion types and recruits agents along each type’s adoption frontier. We evaluate this quantitatively in the counterfactual estimation (Section 7).

Marginal distortions A marginal distortion of perceived benefits scales the relative fuel price the agent perceives. Adopters underuse the technology, and WTP falls by more than proportional scaling. Both main channels persist: the marginal adopter uses more, and demand elasticity rises—though the latter now requires an increasing hazard rate (IHR), rather than only $|\mathcal{E}|$ increasing in price (Appendix X).

Negative correlation between private and social benefits Channel 1 reverses when the conditional external-benefit profile $M(v)$ is decreasing over the relevant range: for example, status buyers who rarely drive adopt an EV first, while commuters who would displace the most gasoline do not. At the margin the distortion then excludes adopters with the greatest abatement, so the purchase-subsidy margin generates a *lower* external benefit than in the undistorted market (??). The elasticity channel can still favor purchase subsidies, so the net effect is empirical. Section 7 examines it in our context, and Appendix ?? derives the reversal formally.

3 Study setting and experimental design

More than two billion people currently lack modern cooking technologies (IEA, 2024), and 73% of African households use charcoal or firewood as their primary cooking fuel (Berkouwer and Dean, 2026c). At the same time, more than two billion people have gained access to electricity in the past two decades (World Bank, 2024), generating widespread policy interest in ‘leapfrogging’ from biomass to electric cooking. East Africa is at the forefront: Kenya’s electric utility launched the “Pika na Power” campaign in 2017, and the government’s National Electric Cooking Strategy Action Plan followed with VAT waivers, support for consumer financing, and a dedicated e-cooking tariff

Figure 3: Charcoal stove and induction stove



Notes: On the left is a charcoal stove instrumented with a high-frequency temperature sensor. On the right is the induction stove we study with the two pots and the frying pan that are included in the purchase bundle.

(Section IV.1). Electric cooking is a particularly effective climate mitigation strategy where grids are clean: renewable sources generated 85% of Kenya’s annual on-grid electricity in 2023 (Kenya Power, 2023).

64% of Kenyan households use firewood or charcoal as their primary cooking fuel (KNBS, 2024). The most common biomass stove in Kenya is a jiko, shown on the left in Figure 3. As a back-of-the-envelope calculation, at baseline respondents use on average 3.1 kilograms of charcoal per day, the production and burning of which emits approximately 4.9 tCO_{2e} per year (Table A1 presents household characteristics; Section 3.5 discusses the conversion from kilograms of charcoal to tons of CO_{2e} emitted).

Cooking-related emissions are substantial. The annual emissions of one cookstove in Kenya are roughly the same as the average annual emissions of a U.S. passenger vehicle, which is driven 11,500 miles per year at 22.2 miles per gallon of gasoline, emitting 8.9 kilograms of CO_{2e} per gallon or 4.6 tons of CO_{2e} per year (EPA, 2025).

3.1 The ECOA induction stove

The right panel in Figure 3 shows the induction stove we study, manufactured by ECOA in Ruiru, Kenya. The induction stove has a power rating of 200W–2000W depending on the desired functionality (from low simmer to high boil) and a voltage rating of 240–250VAC and 50–60Hz. The median ECOA induction stove is in use for 2.2 years, though technology abandonment could be due to a combination of factors, including not only breakage but also non-payment, which shuts down the stove remotely (Figure B2 shows the survival curve).

At the time of our study launch in August 2025, more than 50,000 induction stoves had been sold, primarily through women’s groups and pilot programs implemented through their sales agent programs in Tanzania and Kenya. That said, as the stoves were not yet available in retail outlets in our study area, our study population was generally not familiar with the product. The stove with one pot was priced at US\$72 in Kenya. We offer participants a bundle consisting of the stove plus two pots and one frying pan, priced at US\$84. In this context the price equals the production cost,

so throughout the paper US\$84 is both the stove price p and the cost of the technology.

Domestic residential customers in Kenya face an increasing block tariff. At the start of our study, customers consuming on average less than 30 kWh, 30–100 kWh, or more than 100 kWh per month, respectively, paid US\$0.166, US\$0.20, or US\$0.22 per kWh.¹¹ In our sample, 86% consume less than 30 kWh per month and 99% consume less than 100 kWh per month. We use the lowest tariff band in calculations; the resulting estimates align well with estimates that use self-reported electricity spending data.

The company uses relatively sophisticated credit technologies to help alleviate credit market failures (Section IV.2 provides more detail). To cover these administrative costs, buyers paying in installments under the standard commercial plan must pay back 175% of the principal: an initial down payment of US\$28 is followed by six monthly payments of US\$16 each for a total price of US\$126. Respondents could repay their loans early if they wished, but this would not reduce the total payment amount. The payment structure corresponds to an annualized percentage rate (APR) of 224%, well above Kenya’s central bank rate of 9.75% during the experiment (CBK, 2026).

3.2 Data collection

We implement research activities in peri-urban neighborhoods around the cities of Naivasha, Gilgil, and Nakuru in Nakuru county in Kenya. Nakuru County is relatively representative of Kenya in terms of its fuel mix: 63% of households cook with firewood or charcoal and 65% have access to electricity, compared with 64% and 58% nationwide, respectively (KNBS, 2024). Respondents are visited up to four times (Figure A3 presents a timeline).

Baseline survey Study enumerators enrolled 2,503 people while walking around the study neighborhoods (Table A1 presents summary statistics). During enrollment, enumerators asked a small number of demographic and socioeconomic questions. To qualify for study participation, respondents had to own a standard Kenyan stove (shown on the left in Figure 3); spend at least half of their cooking fuel budget, and at least US\$4 per week, on charcoal; be at least 18 years of age; have a pre-paid Kenya Power electricity meter in their homes; have a household size of at least 3 individuals; have basic literacy; and not currently own an electric or energy efficient charcoal stove.

Enumerators also installed temperature sensors on households’ charcoal stoves. These would continue collecting 2-minute interval data continuously (Section IV.3 provides more detail).

Main survey 2,134 respondents completed the main survey, which was conducted around 50 days after the baseline visit (since attrition at this stage occurred before randomized assignment was revealed, it should not affect outcomes). Enumerators implemented the random treatments, provided (randomized) information about the induction stove, and elicited willingness-to-pay (WTP) using a Becker et al. (1964) mechanism (Section IV.4 provides more detail).

¹¹Kenya’s electricity tariff consists of a fixed component plus a floating component pegged to international fuel prices. The total price per kWh for customers in the top band fluctuated between US\$0.2129 and US\$0.2219 over the eight-month study duration.

Respondents whose WTP was at least their randomly assigned price were expected to purchase the induction stove during the main visit (98% complied). They were then offered a standard 30-minute training on how to use the stove and how to contact customer service representatives in their area. Upon installation, the electric stoves record power usage in kilowatt-hours (kWh) on a 15-minute interval basis. This is transmitted wirelessly to remote servers in real-time, allowing us to subsidize the electricity used for the induction stove. The household surveys also ask about electricity spending, which we convert to usage using the US\$0.166 per kWh electricity tariff.

Endline surveys 2,081 respondents completed the endline survey, conducted on average 40 days after the main survey.¹² Enumerators also briefly visited respondents around six months after the initial sensor deployment to download temperature data.

Loan repayment For respondents who paid for the stove in installments, we observe daily payment data. In addition to measuring repayment, this also reveals the days on which the respondent is ‘locked out’ of their stove: this happens if they fall behind on their payments by at least one week (Section IV.2 provides more detail).

Market price survey We collect a panel dataset of charcoal prices by conducting market surveys over the course of the experiment across all three cities. This allows us to convert household charcoal expenditure data to kilograms of charcoal purchases.

3.3 Randomized treatments

Between the baseline and main surveys, each respondent is assigned to several randomized treatments (Figure A4 presents the randomization tree). All treatments are fully cross-randomized and stratified. Treatment assignment is furthermore stratified by baseline LPG ownership, baseline electricity expenditure, and baseline charcoal expenditure, and is balanced on observable baseline characteristics (Table B1).

Fixed-cost subsidy Participants were cross-randomized into different subsidy amounts for the fixed cost of buying the induction stove bundle, which included the stove and two pots and frying pan, shown in Figure 3. 49% of participants received a discount of 10% (for a price of US\$73) and 46% received a discount of 75% (for a price of US\$20) from the US\$84 price.¹³

Marginal-cost subsidy Participants in the usage subsidy treatment groups receive a subsidy on the cost of the electricity that is used to operate the induction stove. One-third of participants

¹²Stove adopters are 2.2 percentage points less likely to attrit, but attrition is otherwise balanced by random treatment assignment (Table B2). 68% of attriters had moved outside the study area (Table B3).

¹³The remainder of participants had discounts of 46% (0.2% of participants), 50% (2.5%), or 69% (1.8%). All prices had positive probability of being selected from the feasible range of US\$20 (since a down payment of US\$12 plus at least some repayment was required by the stove firm) to US\$73 (for ethical reasons all study participants received a discount of at least 10%).

do not receive a marginal-cost subsidy, one-third receive a subsidy of 25%, and one-third receive a subsidy of 75%. During the main visit, prior to the WTP elicitation, recipients are informed of the level and duration of subsidy they would receive and that only electricity used by the electric stove is subsidized. For a sense of magnitude, given realized usage averaging 15 kWh per month at a tariff of US\$0.166, total marginal-cost subsidy payouts over the six-month subsidy period averaged US\$3.7 and US\$11 for respondents in the 25% and 75% subsidy groups, respectively.

Subsidies are paid out weekly over SMS using a unique, non-transferable token ID ([Section IV.5](#) provides more detail). We designed this intentionally to mirror how households normally pay for electricity. Households normally buy tokens using a mobile phone, and then enter the token ID into their prepaid meter. 30% of participants normally do this once per week, and 66% buy tokens every 3–15 days. This method also matches how the electric utility normally implements subsidy programs. During the endline survey, 86% of respondents reported that they had been receiving the SMSes containing the token IDs, and 96% of those reported that they had been successfully able to enter the tokens into their electricity meters.

Payment structure One half of participants were randomly assigned to pay up front and one half to pay with an interest-free loan. Participants paying up front had to pay the full (randomly assigned) price during the main visit. Participants paying with a loan had to pay a down payment of US\$12.¹⁴

For those paying with a loan, starting one month after the purchase date the total amount due increases daily for 90 days. If a respondent did not meet the amount due, the stove would shut off seven days later until they made another payment, at which point the amount due would be recalculated from the new payment date ([Section IV.2](#) provides more detail). Respondents could pay early but this would not reduce the total amount owed.

Information treatment All respondents receive two information sheets that show what meals they can cook with a charcoal stove and with an electric stove ([Figure A5](#)). 86% of participants are assigned to an information treatment group: their sheets list the cost of the electricity used to cook various meals on an induction stove and on a charcoal stove. For participants in the usage subsidy groups, the information sheets show costs with and without subsidy amounts.

3.4 Relevance to the model

The context described above closely matches the model presented in [Section 2](#). To start, we observe an extremely inelastic intensive margin. As [Section 4.4](#) will discuss, we observe no statistically significant drop in total time spent cooking (if anything, the point estimate is a 9% increase): the 25 minute per day decrease in charcoal cooking time is offset by a 37 minute increase in electric cooking time with very little simultaneous stove use (columns 7–9 of [Table A2](#)).

¹⁴On the first two days of the main survey, September 23 and 24, a down payment of US\$20 was used due to a miscommunication with the study partner. The exclusion of surveys collected on these days (which comprise 5% of the sample) does not meaningfully affect outcomes.

3.5 Converting energy usage to CO_{2e} emissions

Charcoal We assume a fraction of non-renewable biomass (fNRB)—the fraction of used biomass that is permanently removed, as opposed to forest that regrows naturally—of 38%, reflecting the most recently modeled fuelwood estimates for Kenya (Ghilardi and Bailis, 2024; UNFCCC, 2023). Still, our conclusions and results are relatively stable even when assuming an fNRB of 10% or 100% (Section 5.1). We convert estimates of production, processing, and combustion emissions of CO₂, CH₄ and N₂O to CO₂-equivalent emissions assuming a 50-year time horizon. Section IV.6 provides more detail.

Electricity Renewable sources of energy generate 85% of Kenya’s annual grid electricity (Kenya Power, 2023), including geothermal (45%), hydroelectric (19%), and wind (17%). To convert the remaining 15% of generation to CO_{2e} emissions we assume an emissions factor of 1 tCO_{2e} per MWh of electricity, resulting in an average emissions intensity of 154 gCO_{2e} per kWh. While fossil fuel generation meets short-term demand, capacity expansions in recent years have all been renewable: average grid emissions are therefore a relevant margin in the clean cooking transition. We also report outcomes if Kenya’s grid had the average emissions intensity of the U.S. grid.

Liquefied Petroleum Gas We convert LPG expenditures to cylinders purchased using local prices, and convert cylinders to CO_{2e} emissions using a conversion ratio of 4.6 kilograms of CO_{2e} per kilogram of LPG refill.¹⁵

Firewood 17% of households report ever using firewood to cook, but we do not track the quantity of wood collected nor firewood prices. Due to uncertainty in its origin, we refrain from converting the reduction in firewood usage to CO_{2e} emissions or estimating any CO_{2e} abatement generated by fuelwood reductions. Including this abatement would make the cost estimates more favorable.

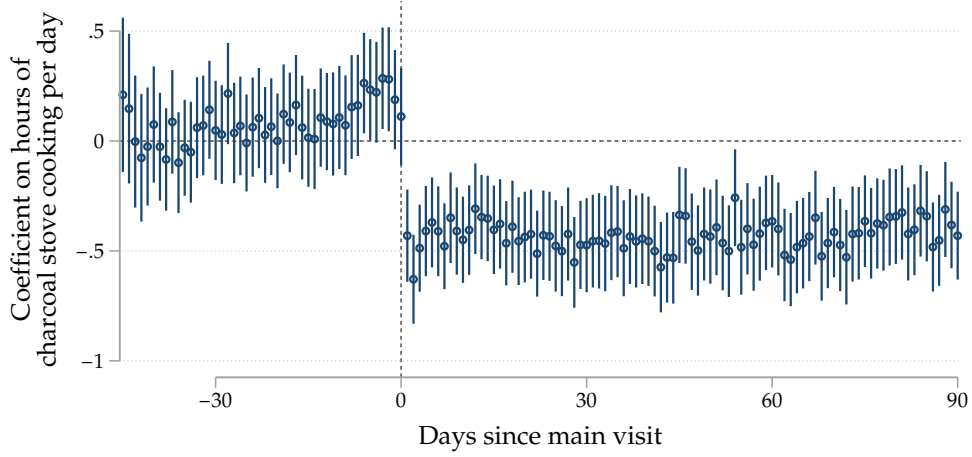
4 Private and social impacts of electric stoves

Of the 2,134 respondents who successfully completed the main visit, 626 bought the induction stove, including 421 with a loan. Since stove purchasing itself was not randomly assigned, we use the randomly assigned subsidy, the randomly assigned loan access, and their interaction as instrumental variables to estimate the causal impacts of induction stove adoption on private and social outcomes (Section V presents these outcomes in the context of the pre-analysis plan). This consistently yields an *F*-statistic above 40.¹⁶

¹⁵One refill of a 6 kg canister of LPG in East Africa emits 27.8 kg CO_{2e}, or 4.6 kg CO_{2e} per kg LPG (USAID, 2021). This includes 3.0 kg CO_{2e} during combustion, 1.2 kg CO_{2e} during production, and 0.4 kg CO_{2e} during transport and raw materials production. A typical 6 kg LPG canister in Kenya costs approximately Ksh 1,400.

¹⁶56% of respondents with a high subsidy and access to a loan bought the stove, whereas 1% of respondents with a low subsidy and paying up front bought the stove. 23% of respondents with a low subsidy and access to a loan bought the stove. 39% of respondents with a high subsidy but no access to a loan bought the stove.

Figure 4: Charcoal stove usage from Stove Use Monitors



Notes: Coefficients from a fixed effects regression using hourly data in which the outcome variable is the fraction of that hour the respondent was cooking (derived from the temperature data using the detection algorithm described in [Section IV.3](#)), multiplied by 24 for ease of interpretation. The regression includes date, event date, and hour of day fixed effects. Standard errors are clustered by respondent. Estimates from regressions with data collapsed to pre- and post-period differences in [Table A2](#).

4.1 Charcoal stove usage

We convert temperature sensor data to an hourly panel dataset of cooking time by respondent, spanning from 45 days before to 90 days after the main visit. [Figure 4](#) plots estimates of separate coefficients of the daily difference between adopters and non-adopters from 45 days before the main visit until 90 days after the main visit. Column 1 of panel a in [Table 2](#) shows that daily time spent cooking using a charcoal stove, as measured by the temperature sensors, decreases by 31 minutes per day ([Table A2](#) shows additional robustness checks). This corresponds to a 34% reduction in daily cooking time caused by adoption of the electric stove.

4.2 Energy expenditures

[Table 2](#) presents estimates of the causal effect of stove adoption on energy expenditures. Self-reported monthly charcoal expenditures, shown in column 2 of panel a, decrease by US\$10 (66%). This is estimated from all compliers in the instrumental variables approach. Thus, it factors in behaviors like ‘stacking’ (where households continue to use their old stoves) as well as households choosing not to use their stoves. Without those behaviors, the estimated charcoal usage reduction would likely have been higher.

Column 5 shows that self-reported monthly electricity expenditures increase by US\$2.3 (65%) per month.¹⁷ This is very close to the value of the induction stoves’ average consumption of 15 kWh per month, which equates to US\$2.5 per month at the Kenya Power tariff of US\$0.166 per kWh.

¹⁷To address the confounding effect of the electricity subsidies, column 5 defines total electricity spending as self-reported electricity spending plus subsidy receipts. Column 6 estimates self-reported electricity spending among the subsidy control group.

Table 2: Impact of induction stove purchase (instrumental variables)

(a) Energy usage

	Hours per day cooking with charcoal	Monthly expenditures (USD)					
		Charcoal	LPG	Wood	Electricity		Total
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Bought induction stove (=1)	-0.5*** (0.2)	-10.0*** (0.8)	-0.9 (0.6)	-0.3 (0.2)	2.3*** (0.4)	2.1*** (0.7)	-8.9*** (1.1)
Observations	5301720	2081	2081	2081	2081	703	2081
Control Mean	1.5	15.1	5.5	0.7	3.5	3.7	24.9
F-Stat	130	241	241	241	241	76	241

(b) Emissions (tons of CO_{2e} per year)

	Charcoal	LPG	Electricity	Total
	(1)	(2)	(3)	(4)
Bought induction stove (=1)	-2.50*** (0.20)	-0.03 (0.02)	0.02*** (0.00)	-2.50*** (0.20)
Observations	2081	2081	2081	2081
Control Mean	3.77	0.17	0.04	3.99
F-Stat	241.34	241.34	241.34	241.34

Notes: Instrumental variables estimates use the randomized treatment price, the randomized credit condition, and their interaction as exogenous instruments. LPG is liquefied petroleum gas. Panel a shows the impact of induction stove purchasing on daily cooking and monthly energy expenditures. Column 1 uses hourly temperature sensor data (standard errors clustered by respondent; additional specifications in [Table A2](#)). Column 5 adds subsidies to electricity expenditure. Column 6 estimates self-reported electricity spending among the subsidy control group. Panel b shows the impact on emissions. [Section 3.5](#) describes the methodology for converting energy expenditures to emissions. ‘Control mean’ evaluated among non-adopters in the complier sample.

There are additional, smaller decreases in wood and LPG expenditure. Aggregate monthly energy expenditures decrease by US\$8.9 (36%). Assuming a 10% annual discount rate, this corresponds to US\$209 over the stove’s 2.2 year expected usage duration.

At the stove’s price of US\$84, the savings generate a return on investment of 114% per year. When paying with the commercial installment plan described in [Section 3.1](#) the return is 80%. While these returns are substantial, they are less than the payment plan’s 224% APR. For most households, the monthly energy savings do not cover the monthly repayment requirements, rendering adoption with credit prohibitively expensive for many. For comparison, the high and low subsidy pay plans in the experiment required monthly payments of US\$3 and US\$21, respectively.

The only baseline socioeconomic characteristic along which we see treatment effect heterogeneity is ownership of an LPG stove: non-owners save US\$12 per month while owners save US\$8 per month ([Table B10](#)). The p-value for this comparison is 0.095 but does not survive a multiple hypothesis testing adjustment.

4.3 Emissions

We convert energy expenditures to CO₂e emissions using the methodology described in [Section 3.5](#). While electricity expenditures constitute approximately 36% of total household energy expenditure among those who bought the stove, they constitute less than 3% of their baseline household emissions.¹⁸ The increase in electricity usage thus generates grid emissions of 0.02 tCO₂e per year. For comparison, annual emissions from charcoal cooking average 4.9 tCO₂e and from LPG cooking average 0.3 tCO₂e ([Table A1](#)). Induction stoves as a technology are highly efficient in converting energy to heat. While Kenya’s 85% renewable grid contributes to the low emissions, even at U.S. average grid emissions intensity the emissions from electric cooking would only amount to 0.05 tCO₂e per year.

Panel b of [Table 2](#) shows an annual emissions reduction of 2.5 tCO₂e. This is effectively equivalent to the reduction in charcoal-related emissions, since the modest increase in electricity-related emissions is offset by a modest decrease in LPG-related emissions (as discussed in [Section 3.5](#), we do not quantify fuelwood reductions).

4.4 Co-benefits and constraints

We briefly discuss the impacts of electric cooking on diet and health, and how power quality might affect adoption, below. These co-benefits and constraints are important from the perspective of poverty reduction and welfare improvement. However, they only affect the subsidy cost per tCO₂e to the extent that they move WTP and usage directly, which our measurements already incorporate. They matter for a social planner whose decisions incorporate welfare gains; however, even then they remain difficult to quantify. We therefore do not include them in the welfare calculations. Berkouwer and Dean ([2026c](#)) presents additional detail on the drivers and impacts of electric stove adoption that relate to health, socioeconomic status, and electrical grid quality.

Diet Induction stove adoption generates a 12 percentage point decrease in the fraction of people who cook ugali, mukimo, or matoke (three starch staples) and an 11 percentage point increase in the fraction of people who cook rice (Berkouwer and Dean, [2026c](#)). Anecdotally, rice is easier to cook on an electric stove than other starches. We observe no other meaningful changes in which foods people cook nor the number of unique foods they cook. We do not observe any time savings: the 25 minute per day decrease in charcoal cookstove usage is more than offset by the 37 minutes per day of electric cooking. Given that respondents use both stoves simultaneously only 0.7% of cooking time, this leaves households cooking 10 minutes more per day (columns 7–9 of [Table A2](#)).

Health We see meaningful reductions in self-reported cough and breathlessness among respondents and their children. In a similar context, Berkouwer and Dean ([2026a](#)) found health improvements of similar magnitude. While these are potentially important for welfare, those improvements

¹⁸These numbers exclude transportation-related emissions or other emissions generated outside the home.

did not generate benefits in terms of long-term clinical health or economic outcomes like hours worked in that study.

Power quality In a similar context in Kenya, Wolfram et al. (2027) record an average of 61 minutes of power outage per day, as well as imperfect voltage quality: average voltage of 233V is 3% lower than nominal voltage of 240V. This could have an important dampening effect on the value of electric cooking in this context.

5 Green subsidy efficacy

5.1 Fixed-cost subsidies

We first evaluate the subsidy cost of incentivizing a marginal adoption by leveraging the demand curve elicited through the BDM exercise (panel a of Figure 7). The high price is the competitive market price minus a small subsidy $p - s_l$. A larger subsidy lowers the price further to $p - s_h$, such that $0 \leq s_l < s_h$. Denote q_l and q_h to be the fraction of individuals who would buy the stove with the low or high subsidy, respectively. The subsidy cost required to incentivize one additional adoption can then be denoted as follows:

$$C_s = \frac{\text{Additional subsidy cost}}{\text{Additional adoptions}} = \frac{q_h s_h - q_l s_l}{q_h - q_l} \quad (19)$$

In the context of this study, absent access to a loan 40% of respondents would buy the stove with the high subsidy whereas 2.6% of respondents would buy the stove with the low subsidy. Applying Equation 19, the principal spends $C_s = \text{US\$67}$ in subsidy expenditures to incentivize one additional adoption.

Subsidy efficiency can then be calculated by combining this with the stove price $p = \text{US\$84}$ and the benefits per additional adoption. Marginal buyers of induction stoves in our context save $\text{US\$9.7}$ per month and abate 2.6 tCO_2e per year. Discounting at 10% per year, this gives a discounted lifetime private benefit per marginal adopter of $\hat{b} = \text{US\$227}$ and lifetime abatement of $\hat{\phi} = 5.7 \text{ tCO}_2e$ (the total discounted monthly savings and annual abatement coefficients in columns 8 and 10 of Table 5 after 2.2 years). Using a Social Cost of Carbon (SCC) of $\text{US\$120}$ (the 2020 Environmental Protection Agency estimate using a 2.5% discount rate; EPA, 2023), this corresponds to a marginal externality benefit of $\text{SCC} \cdot \hat{\phi} = \text{US\$688}$.

The causal estimates from the instrumental variables (IV) regression were identified off of the compliers, who have a willingness-to-pay (WTP) between the low and high prices ($p - s_h = \text{US\$20}$ and $p - s_l = \text{US\$73}$). The local average treatment effect is thus estimated for adopters who are marginal to an increase in the subsidy from s_l to s_h . Because compliers in the IV estimation are exactly the adopters marginal to the subsidy increase, we can conduct several relevant policy calculations.

The results below present estimates for the subsidy cost per tCO_2e and the observable welfare

gain per subsidy dollar, reflecting the objective functions of the environmental principal and the social principal, respectively. Table B5 defines and presents results for two additional measures: marginal value of public funds and resource cost per ton of CO₂e abated.

Subsidy cost per tCO₂e We first consider an environmental principal who wishes to maximize CO₂e abatement while minimizing subsidy expenditure.¹⁹ This could be a budget-constrained government agency or a cost-minimizing company or organization that wishes to meet an abatement target. The additional subsidy expenditure required to abate one additional ton of CO₂e is then given by:

$$\text{Subsidy cost per tCO}_2\text{e} = \frac{\text{Subsidy cost per additional adoption}}{\text{Tons of CO}_2\text{e abated per adoption}} \quad (20)$$

Applying the estimates from above, subsidies for induction stoves in this context have a subsidy cost per tCO₂e of US\$12.

Observable welfare gain per subsidy dollar Since WTP is distorted in this context, we cannot use it to measure consumer surplus. We instead evaluate the observable components of welfare per subsidy dollar, setting unobserved switching costs to zero. In the case of a pure allocation distortion, this consists of fuel savings plus the value of the externality avoided minus the production cost of the improved stove (conservatively, we assume the cost of the traditional stove is 0):

$$\text{Benefit per subsidy dollar} = \frac{\text{Benefit per marginal adoption}}{\text{Subsidy cost per marginal adoption}} = \frac{\hat{b} + \text{SCC} \cdot \hat{\phi} - p}{C_s} \quad (21)$$

This yields an observable welfare gain of between US\$9.9 and US\$12.1 per subsidy dollar, consisting of US\$3.4 in private fuel savings and US\$10 in environmental benefit, minus stove and resource costs. The interval reflects uncertainty about whether the part of the distortion that was not addressed through our loan treatment generates resource cost. Assuming most of the distortion is allocative yields an observable welfare gain of US\$12.1 per subsidy dollar. If the wedge is entirely a resource distortion, this lowers the estimate to a welfare gain of US\$9.9 per subsidy dollar.

This subsidy cost per tCO₂e is substantially lower than that of many other green technologies. Panel a of Table 3 summarizes the subsidy cost per tCO₂e estimated in this paper as well as in other papers evaluating energy efficient technologies in contexts where adopters likely face significant demand distortions. Panel b lists estimates of green technologies receiving subsidies in the U.S. By and large, subsidies for green technologies in low- and middle-income contexts generate larger social benefits per subsidy dollar spent than similar subsidies for green household technologies in high-income contexts.

These differences are partly driven by the underlying technologies. For example, each US\$84 stove abates on average 2.5 tCO₂e per year while a US\$40,000 electric vehicle abates 0.1–1.5 tCO₂e

¹⁹The decision of whether to express efficiency as abatement per dollar or dollars per abatement is arbitrary; we use subsidy cost per tCO₂e to facilitate natural intuition with the abatement cost curve and offset market prices.

Table 3: Efficiency of selected green subsidies

Technology	Observable welfare gain per US\$1 subsidy (US\$)	Subsidy cost per tCO _{2e} (US\$)	Source
<i>(a) Low- and middle-income contexts</i>			
Electric stoves	US\$9.9–US\$12.1	US\$12	<i>This paper</i>
Solar lanterns	12	10	Rom et al. (2023)
Energy efficient motors	4.1	- ^a	Chaurey et al. (2025)
Improved charcoal stoves	322 ^b	6	Berkouwer and Dean (2026b), Hahn et al. (2026)
<i>(b) High-income contexts</i>			
Electric vehicles	0.4	1,356	Hahn et al. (2026)
Electric vehicles	0.3	1,202	Allcott et al. (2026)
Wind production credits	4.9	46	Hahn et al. (2026)
Residential solar	2.9	90	Hahn et al. (2026)
Appliance rebates	0.2	474	Hahn et al. (2026)
Vehicle retirement	0	876	Hahn et al. (2026)
Hybrid vehicles	0	5,940	Hahn et al. (2026)
Weatherization	0	779	Hahn et al. (2026)

Notes: We conservatively assume a social cost of carbon (SCC) of US\$120; Hahn et al. (2026) assume US\$193. ^aChaurey et al. (2025) estimate the impact of disseminating energy efficient motors for free to non-adopters. We were unable to calculate the subsidy cost per tCO_{2e} from the numbers provided in the manuscript. ^bThe US\$322 benefit estimated by Berkouwer and Dean (2026b) results from the low technology cost (US\$40), large emissions reductions (2.1 tCO_{2e} per year), large financial savings (US\$78–114 per year), and a long stove lifetime (2.7 years).

per year relative to the counterfactual (Muehlegger and Rapson, 2023). However, even conditioning on technology, the channels identified in this paper—differences in the marginal adopter and demand elasticity—drive increases in subsidy efficiency. Section 6 first disentangles how the two channels contribute to subsidy efficiency. Section 7 then estimates the model presented in Section 2 to disentangle how much demand distortions per se reduce the subsidy cost per tCO_{2e} relative to estimates from contexts with fewer demand distortions.

Results under alternative assumptions The benchmark scenario evaluated in this paper uses 2.2 years of stove usage (based on the company’s historical data, shown in Figure B2) and uses Kenya’s average grid emissions. Alternatively, assume that the stove’s lifespan is on average 1 year or 3 years. This changes the subsidy cost per tCO_{2e} among respondents who were not also given access to a loan to US\$26 and US\$8.6, respectively (scenarios 1 and 2 of Table A3).

There is significant uncertainty around the fraction of non-renewable biomass (fNRB). Our primary estimates use the 38% rate that Ghilardi and Bailis (2024) estimate using their Modeling Fuelwood Savings Scenarios (MoFuSS) model, per the recommendations of Article 6.4 of the Paris

Agreement. To measure the sensitivity of our results to this parameter, we also calculate costs under 10% and 100% rates. This changes the subsidy cost per tCO_{2e} to US\$14 or US\$8.7, respectively (scenarios 3 and 4 of [Table A3](#)). These differences are relatively minor because most charcoal-related emissions are incurred not during combustion but during production and processing, which are not affected by the assumed fNRB (Floess et al., 2023).

Finally, Kenya’s electricity generation mix is 85% renewable, generating less than half the emissions per kWh of the U.S. grid. However, even if Kenya’s grid emissions were equivalent to the U.S. grid, the subsidy cost per tCO_{2e} would remain essentially unchanged (scenario 5 of [Table A3](#)). Cooking with charcoal is significantly dirtier than cooking with electricity, such that the large reduction in charcoal cooking emissions is only very slightly offset by the increase in electricity cooking emissions (as discussed in [Section 4.3](#)).

5.2 Marginal-cost subsidies

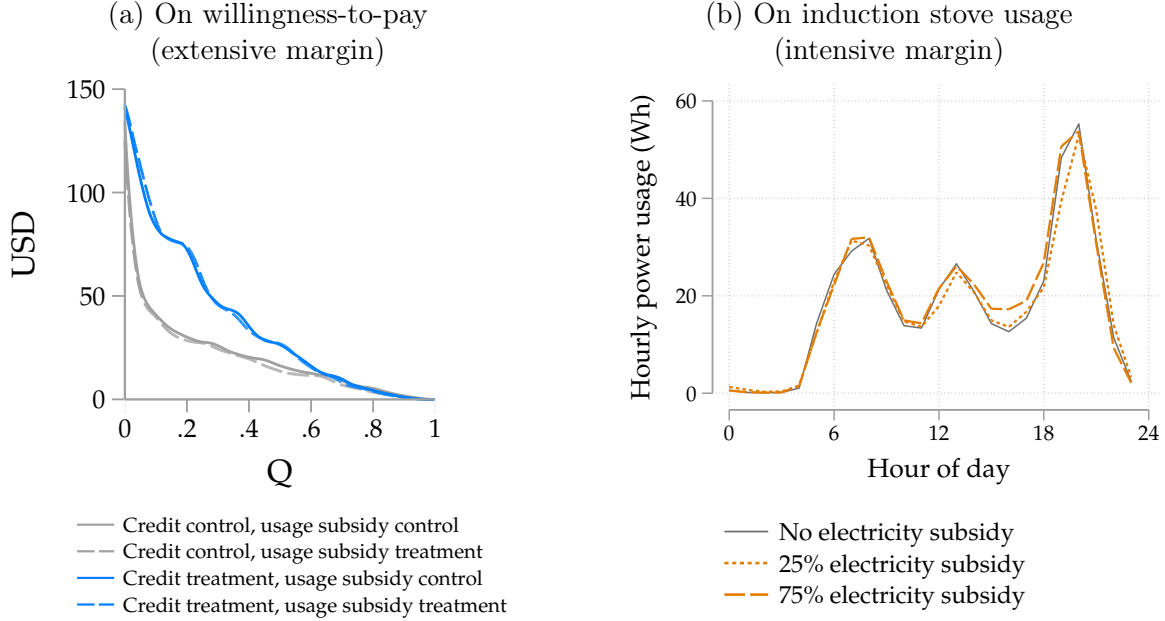
The absolute efficacy of fixed-cost subsidies does not inform their relative efficacy compared with marginal-cost subsidies, which could offer two advantages over a fixed-cost subsidy of equivalent net present value. First, marginal-cost subsidies could encourage selection into adoption by higher users. Second, they could incentivize switching conditional on adoption. Both channels could increase average usage levels conditional on adoption, increasing the efficiency of marginal-cost subsidies in terms of CO_{2e} abated per dollar of subsidy expenditure relative to an equivalent fixed-cost subsidy.

On the other hand, if respondents are myopic or imperfectly attentive to electricity expenditures, they might undervalue electricity subsidies that accrue in the future. This could undermine the efficacy of marginal-cost subsidies. The relative efficacy of these two subsidy instruments is thus an empirical question.

We evaluate the impact of the randomly assigned electricity subsidies, which lowered the cost of cooking by 25% or 75%, each assigned to one-third of the sample. Since we announced the electricity subsidy prior to eliciting WTP, the announcement of the usage subsidy could have affected usage both through a direct effect as well as a selection channel. [Figure 5](#) therefore shows the impact of usage subsidies on both the extensive margin (purchasing the stove) and the intensive margin (using the stove). Panel a shows WTP by whether the respondent was given access to a loan and whether they received a marginal-cost subsidy. The electricity subsidy has no statistically detectable effect on average WTP. We can rule out that each US\$1 in electricity subsidy has more than a US\$0.2 impact (column 3 of [Table A6](#)). Pooled regressions and separate regressions by subsidy level all show no statistically significant impact on WTP ([Table A4](#)). The interaction term between credit and subsidies is positive, but this effect is not significant and very noisy: electricity subsidies may have a positive impact on WTP when respondents’ credit constraints are relaxed, or credit causes selection by people who are slightly more price elastic. We are unable to cleanly evaluate whether this is the case in this context.

This result is what the model predicts at the distortions estimated in this context. On the extensive margin, a marginal-cost subsidy raises perceived WTP at only the pass-through rate

Figure 5: Impacts of marginal-cost subsidies



Notes: Panel a shows the impact of access to credit and electricity subsidies on demand, with demand curves constructed using WTP values elicited through the Becker-DeGroot-Marschak mechanism (point estimates in Table A4). In panel b, electric stove usage in Watt-hours recorded by the induction stoves, separated by whether the respondent was in the electricity subsidy control group, the 25% electricity subsidy group, or the 75% electricity subsidy group (point estimates in Table A6). Figure B3 presents demand curves that adjust for observed repayment rates.

$1 - \gamma$. Taking the ratio of WTP to discounted fuel savings, we estimate a demand distortion of 0.91 (Section 6). Each US\$1 of electricity subsidy should raise WTP by only US\$0.09. This predicted pass-through lies inside our confidence interval, while undistorted one-for-one pass-through is rejected. Viewed through the model, the unresponsiveness of WTP to electricity subsidies is an implication of the same distortion between WTP and fuel savings.

Panel b of Figure 5 presents average hourly induction stove usage by electricity subsidy treatment group. We estimate a demand elasticity of -0.03, and we can rule out that a 1% increase in price causes more than a 0.12% decrease in usage (column 6 of Table A6). The point estimate for the 75% subsidy is larger than the point estimate for the 25% subsidy but less than 10% of mean usage and not statistically different from zero.

Together, these results indicate that fixed-cost subsidy instruments are more effective in this context than marginal-cost subsidies.

6 Demand distortions and subsidy efficacy

One of this paper's central findings is that the relatively low subsidy cost per tCO_{2e} discussed above is caused at least in part by demand distortions. Section 4 showed that the stove's average discounted fuel savings are $b = \text{US\$209}$ after 2.2 years. Respondents paying up front have an average willingness-to-pay (WTP) of US\$20. Defining WTP as $wtp = b \cdot (1 - \gamma)$, this suggests a demand

distortion of $\gamma = 0.91$. Throughout, we compute γ using the pooled savings estimate (Table 2): the wedge is a property of decision-making, and defining it against a common benchmark keeps the two credit arms comparable. The observed WTP relative to the stream of savings corresponds to an implicit discount factor of 0.544 per month, or 0.0007 per year. Put differently, respondent decision-making reflects indifference between receiving US\$1 today or receiving US\$1,487 one year from today.

Figure 5 shows that access to credit in this context increases WTP by 80% (column 4 of Table A4). Respondents given access to a loan have an average WTP of US\$35 suggesting a demand distortion of $\gamma_1 = 0.83$. By the end of data tracking, 271 days after stove sale on average, respondents who had bought the stove with a loan had paid on average 86% of the price (Figure A6). Accounting for observed repayment rates either increases or decreases the treatment effect in percentage terms depending on which outcome is used, but the effect is always large and statistically significant (Table B4). The discount factor increases to 0.0308, at which point US\$1 today is worth US\$32 in one year. The difference between these values illustrates the degree to which capital constraints distort decision-making in this context. We are agnostic as to what drives the remaining gap between fuel savings and WTP: this could be allocative distortions like inattention or misinformation or resource distortions like transaction costs. We therefore bound the welfare effects by assuming either that this entire gap is either entirely an allocation distortion or entirely a resource distortion. The remaining gap may also reflect preferences that our survey questions do not capture: while this would affect the welfare estimates, it would not change the effect of the distortion on the subsidy cost per ton of CO_{2e} abated.

How do these demand distortions affect subsidy efficacy? Table 4 presents estimates of subsidy efficiency for respondents whose demand was more distorted and less distorted. Its rows are the empirical counterparts of the model’s summary statistics. The Channel 1 rows are the empirical counterparts of the average usage of induced adopters ($\bar{u}$), valued at the per-unit fuel saving and at the per-unit externality. The Channel 2 rows are the empirical counterparts of demand elasticity and the number of subsidy recipients per induced adoption (ζ). Taken together, the subsidy cost per ton of CO_{2e} abated is US\$20 among the group with access to a loan, whereas it was US\$12 among those with a higher demand distortion. The distortion thus decreases the subsidy cost of abatement by 41%.

The social principal’s observable welfare gain consists of fuel savings, externality benefit, stove production cost, as well as any resource cost incurred. This therefore depends on the degree to which the distortion is an allocation distortion or a resource distortion. To start, we assume the difference in WTP between the two groups is a resource distortion with rate $1 + r_c = \frac{1-\gamma_1}{1-\gamma_0}$, or $r_c = 80\%$. Reassuringly, this is close to the rate the market charges: the commercial installment product for this device requires repaying 175% of the amount borrowed. The composition of the wedge that remains is unknown. If the remaining wedge is entirely allocation, respondents with loan access carry $r = 0$ (the study loans were interest-free) and respondents paying up front carry r_c . If the remaining wedge is entirely resource, the rates compound to $\frac{\gamma_1}{1-\gamma_1} = 494\%$ for respondents with

Table 4: Impact of demand distortions on subsidy efficacy through the two channels

	Less distorted ($\gamma_1 = 0.83$)	More distorted ($\gamma_0 = 0.91$)	Advantage of subsidies in a distorted market
<i>Channel 1: Marginal adopter</i>			
Fuel savings per marginal adoption ($\hat{b}$)	US\$175	US\$227	+30%
Abatement per marginal adoption ($\hat{\phi}$)	4.8 tCO ₂ e	5.7 tCO ₂ e	+19%
<i>Channel 2: Subsidy cost</i>			
Extensive margin elasticity to fixed subsidy	$\mathcal{E} = -0.9$	$\mathcal{E} = -2.3$	
Subsidy expenditure inframarginal	38%	6.5%	
Subsidy expenditure per marginal adoption	US\$96	US\$67	-29%
<i>Aggregate difference</i>			
Subsidy cost per tCO ₂ e	US\$20	US\$12	-41%
Observable welfare gain per subsidy dollar			
Remaining wedge is 100% allocation	US\$7.0	US\$12.1	+73%
Remaining wedge is 100% resource	US\$7.6	US\$9.9	+30%

Notes: How demand distortions affect subsidy efficacy. ‘Less distorted’ are respondents with access to an interest-free loan. ‘More distorted’ are respondents who pay the full price up front. Fuel savings and abatement per marginal adoption are totals after 2.2 years. Observable welfare gain consists of fuel savings plus externality benefit minus the economic cost of stove manufacturing and the resource cost (if any). Under an allocation distortion the wedge consumes nothing; under a resource distortion each induced adopter incurs additional resource cost but the subsidy lowers the resource cost for inframarginal borrowers. The wedge attributable to credit constraints is a resource distortion with rate $r_c = 80\%$ (Section 6). The two welfare rows correspond to the two extreme compositions of the remaining wedge γ_1 , which both groups face: if it is entirely allocation, the loan group carries $r = 0$ and the cash group $r = r_c$; if it is entirely resource, the loan group carries $r = \frac{\gamma_1}{1-\gamma_1} = 494\%$ and the cash group $r = \frac{\gamma_0}{1-\gamma_0} = 973\%$. Section 2.2 describes the theoretical underpinnings of channels 1 and 2. Section 6.1 and Section 6.2 present empirical estimates for channels 1 and 2. The GMM estimation in Section 7 uses the $\hat{b}$ and $\hat{\phi}$ estimates.

loan access and $\frac{\gamma_0}{1-\gamma_0} = 973\%$ for respondents paying up front. Since $\mathbb{C}_s > p$ among respondents with loan access, a resource wedge causes subsidies to generate a larger welfare gain than an allocation wedge in that group (denoted $s_1\zeta(r) > p$ in Section 2.2). Among respondents paying up front $\mathbb{C}_s < p$, so the same reclassification lowers their welfare gain.

Section 6.1 and Section 6.2 quantify how the externality generated by the marginal adopter (channel 1) and demand elasticity (channel 2), respectively, drive these results.

6.1 Channel 1: Fuel reduction of marginal adopters

We first test the impact of demand distortions on the average externality of the marginal adopter. Throughout, we define a marginal adopter as having a WTP between the low price and the high price, so that they would adopt the stove if and only if there was an increase in the subsidy: $p - s_h \leq wtp < p - s_l$. There are 387 marginal adopters in the credit control group and 378 marginal adopters in the credit treatment group for a total sample of 765 marginal adopters.

In the context of the model, when agents face larger demand distortions, agents who are marginal

Table 5: Monthly savings and annual abatement among marginal electric stove adopters

	Charcoal				Electricity		Total			
	(1) USD	(2) USD	(3) CO _{2e}	(4) CO _{2e}	(5) USD	(6) USD	(7) USD	(8) USD	(9) CO _{2e}	(10) CO _{2e}
Credit (=1)	-1.0 (1.4)	-1.0 (1.4)	-0.2 (0.4)	-0.2 (0.4)	0.1 (0.8)	0.1 (0.8)	-0.7 (2.0)	-0.7 (2.0)	-0.2 (0.4)	-0.2 (0.4)
Bought stove (=1)	-10.5*** (0.6)		-2.6*** (0.2)		2.3*** (0.3)		-9.7*** (0.9)		-2.6*** (0.2)	
Bought X No credit		-10.5*** (0.6)		-2.6*** (0.2)		2.3*** (0.3)		-9.7*** (0.9)		-2.6*** (0.2)
Bought X Credit	1.6* (0.9)	-8.9*** (0.6)	0.4* (0.2)	-2.2*** (0.2)	-0.2 (0.5)	2.2*** (0.3)	2.2* (1.2)	-7.4*** (0.8)	0.4* (0.2)	-2.2*** (0.2)
Observations	754	754	754	754	754	754	754	754	754	754
Control Mean	15.7	15.7	3.9	3.9	3.6	3.6	25.5	25.5	4.1	4.1

Notes: Instrumental variables regressions using the randomly assigned price as an instrument for adoption to estimate its impact on spending and emissions among marginal adopters (i.e., $WTP \in [20, 73]$ US\$). USD amounts are monthly; CO_{2e} amounts are annual. IV compliers correspond to marginal adopters. Odd columns test whether the marginal adopters' average externality differs by credit condition. Even columns estimate the positive externality by credit condition. Endline surveys were conducted with 754 of the 765 marginal adopters (98.6%).

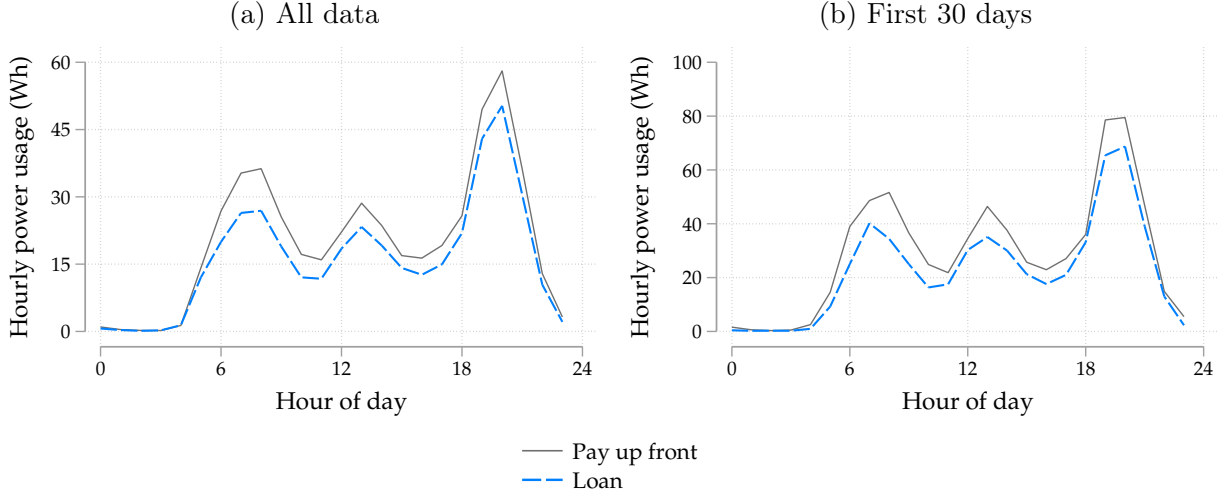
to a subsidy should unambiguously have higher average usage, and hence higher fuel savings (Section 2.1.1). Columns 7 and 8 of Table 5 offer a test of this. Marginal adopters with a larger demand distortion generate on average US\$227 in discounted private savings over 2.2 years, whereas adopters whose demand distortion was reduced through access to credit generate US\$175, a statistically significant difference.

Since social and private benefits are positively correlated in this context, this translates into a similar difference in positive externalities. Columns 9 and 10 of Table 5 show that this is the case. Marginal adopters in the credit control group abate 5.7 tCO_{2e} after 2.2 years whereas marginal adopters in the credit treatment group abate 4.8 tCO_{2e}. In other words, the credit constraints in this context increase the abatement caused by marginal adopters by 19%.

Figure 6 plots induction stove usage over the course of the day among adopters who were marginal to the subsidy, by credit treatment group. Marginal adopters in the credit control group have an average usage of 538 Wh per day, which is 18% higher than marginal adopters in the credit treatment group, who have an average daily usage of 455 Wh per day (Table A5 shows the estimates corresponding to panel a).

This difference could reflect a treatment effect rather than selection. For example, having to pay back a loan leaves less money available for electricity expenditures. However, panel b shows that these effects begin within the first month, when adopters paying up front would have just faced a large direct liquidity shock from adoption. If anything, this would bias the estimates toward finding less usage among adopters paying up front. The difference in usage across the two groups is therefore likely to be a result purely of selection rather than a treatment effect. Either way, the interpretation of this wedge does not affect the subsidy efficacy calculations.

Figure 6: Electric stove usage among marginal adopters by credit condition



Notes: Hourly usage in Watt-hours by randomized credit condition among marginal adopters, defined as having $p - s_h \leq wtp_i < p - s_l$. Table A5 presents point estimates, which are significant with $p < 0.05$.

6.2 Channel 2: Subsidy expenditure per marginal adoption

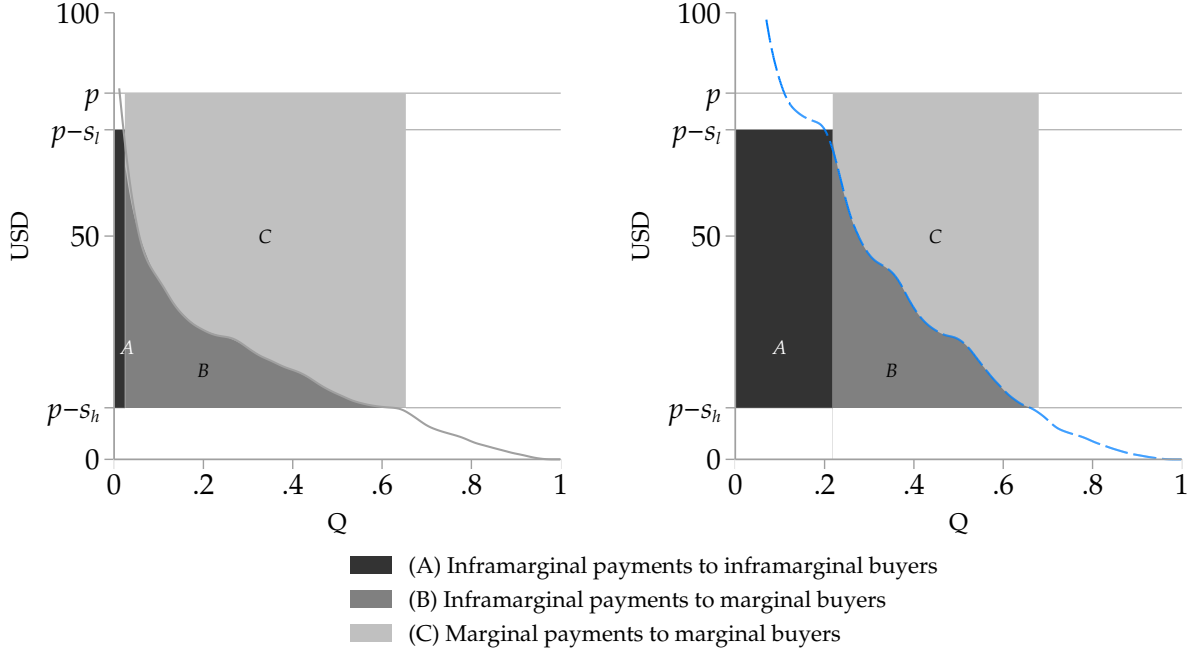
The second channel through which demand distortions increase subsidy efficacy is by increasing demand elasticity, which both increases the number of agents who are marginal to the subsidy, and reduces the number of inframarginal payments. Distorted elasticity at any price equals undistorted elasticity at the scaled-up price, so the inframarginal cost per induced adoption falls with the distortion (Section 2.1.2). The cost increase resulting from a subsidy increase consists of subsidy expenditures towards both inframarginal and marginal buyers:

$$q_h s_h - q_l s_l = \underbrace{q_l (s_h - s_l)}_{\text{Increased payouts to inframarginal buyers}} + \underbrace{s_h (q_h - q_l)}_{\text{Full payouts to marginal buyers}} \quad (22)$$

Figure 7 presents intuition on how these effects operate in the context of the demand curves elicited through the BDM. Area A represents the first term in Equation 22 and $B \cup C$ represents the second term. Area A as a fraction of areas $A \cup B \cup C$ denotes the fraction of additional subsidy expenditures disbursed to inframarginal buyers. Comparing panels a and b, demand distortions in this context both decrease payouts to inframarginal buyers in absolute terms and as a fraction of total payouts. 2.6% of people in the credit control group would buy the stove at the high price whereas 21% of people in the credit treatment group would. As a result, among the credit treatment group 38% of subsidy expenditures flow to inframarginal buyers whereas in the credit control group only 6.5% of expenditures do.

Taken together, the environmental principal incurs a total subsidy cost of US\$96 per additional sale among respondents with access to a loan, whereas respondents in the credit control group require additional subsidy spending of US\$67 per additional sale. In other words, relaxing demand distortions increases the subsidy cost of achieving one additional sale by 42%.

Figure 7: Marginal and inframarginal expenditures by credit condition
(a) More distorted demand (b) Less distorted demand



Notes: Demand curves constructed using WTP values elicited through the BDM mechanism. Respondents having to pay up front have more distorted demand than respondents randomly given access to a loan. Demand distortions increase the proportion of marginal buyers and decrease the proportion of inframarginal buyers that result from an increase in the subsidy from s_l to s_h .

These impacts can be summarized by the change in the demand elasticity, which as a percentage term reflects both marginal adopters (in the numerator) and inframarginal adopters (in the denominator). Table 6 estimates the price-quantity coefficients. Column 5 shows a demand elasticity of -2.3 among participants required to pay the entire price up front and column 6 shows -0.9 among people paying with a loan.

An alternative, policy-relevant measure of demand elasticity would be to take s_l as the policy baseline and then to estimate the percentage quantity and price change from this starting point. This results in a demand elasticity of -20 among the control group and -2.3 among respondents whose credit constraints were partially relaxed.

Columns 2 and 3 indicate that the increased elasticity includes an increase in the numerator: there are more marginal adopters per US\$1 subsidy. Given that there are also fewer inframarginal adopters (the denominator in the elasticity calculation), column 5 shows that respondents with a larger demand distortion have an even larger demand elasticity.

7 Model estimation and counterfactuals

Even among respondents who are given access to a loan, willingness-to-pay (WTP) of US\$35 is still well below the total savings of US\$209 after 2.2 years (savings are discounted at 10% per year).

Table 6: Extensive margin demand elasticities by credit treatment group

	Quantity (%)			Log(Quantity)		
	(1)	(2)	(3)	(4)	(5)	(6)
Price (USD)	-0.84*** (0.01)	-0.96*** (0.02)	-0.80*** (0.01)			
Log(Price)				-1.23*** (0.01)	-2.33*** (0.02)	-0.88*** (0.01)
Observations	765	387	378	765	387	378
Sample		C=0	C=1		C=0	C=1

Notes: We estimate elasticities over the policy-relevant sample of compliers, i.e., anyone with willingness-to-pay between $p - s_h$ and $p - s_l$. Column 1 shows that a US\$1 increase in price causes a 0.8 percentage point decrease in the quantity demanded. Columns 2 and 3 show that this decrease is steeper for respondents in the control group than in the treatment group. The lower rate of inframarginal adoption exacerbates the difference in elasticities in columns 5 and 6, as the small denominator among the credit control group causes a larger demand elasticity.

It is reasonable to think that our short intervention did not fully alleviate all demand distortions experienced by households in this context. It may not have fully alleviated credit constraints: given that average monthly savings were US\$8.9 and the loan had a four-month repayment period, someone with a binding savings constraint might only be able to pay US\$36 over the period of the loan. We therefore estimate a generalization of the model specified in [Section 2](#). This allows us to generate counterfactuals to quantify outcomes in alternative scenarios where the demand distortion is alleviated further. Of course, efficient WTP in this context may not be exactly equal to the discounted stream of fuel savings: other factors like risk aversion or higher time discounting could also lower WTP relative to the total stream of savings causing $\gamma > 0$, or the induction stove could generate additional unobservable benefits such that WTP exceeded savings. Still, we think the full stream of financial benefits discounted at 10% per year is a useful benchmark against which to compare the results.

The model clarifies how the demand distortion and the private-social benefit correlation affect subsidy efficacy. We are agnostic about what drives this heterogeneity. For example, in another context a clean technology might generate a reduction in pollution exposure that increases WTP even in the absence of any financial savings: this would still cause a positive correlation between private and social benefits, and the model’s insights would still apply.

We do not take a stance on whether the wedge generates additional costs or not, since this affects the agent’s decision-making but does not have any direct welfare consequences.

7.1 Set-up

We assume agents i are distributed Weibull by their private benefit from adopting: $(b_i/\hat{b}) \sim \text{Weibull}(\Omega_1, \Omega_2)$, with discounted private benefits after 2.2 years $\hat{b} = \text{US\$209}$ (from panel a, column 7 of [Table 2](#)). The Weibull’s shape parameter governs tail thickness, allowing the data to discipline how concentrated WTP is around its mean; for $\Omega_1 \geq 1$ it satisfies the assumed increasing hazard

rate (IHR) condition (the estimate of $\Omega_1 = 1.9$ satisfies this condition).

The model assumed usage was perfectly correlated with fuel savings and emissions, but in practice this correlation is imperfect. We therefore allow the correlation between social benefit and private benefit to be controlled by Ω_3 through $\phi_i = \hat{\phi} \left[\Omega_3 (b_i/\hat{b}) + (1 - \Omega_3) \varepsilon_i \right]$, with $\hat{\phi} = 5.5$ tCO₂e (from panel b, column 4 of Table 2 multiplied by 2.2 years), $\Omega_3 \in [0, 1]$, and ε_i an independent draw from the same Weibull distribution.

For each credit condition $c \in \{0, 1\}$ the mean demand distortion $\gamma_c \equiv \mathbb{E}[\gamma_i|c]$ is defined by $w\bar{t}p_c = (1 - \gamma_c)\hat{b}$, with observed mean WTP $w\bar{t}p_0 = \text{US\$20}$ and $w\bar{t}p_1 = \text{US\$35}$ (from Table A4), giving $\gamma_0 = 0.91$ and $\gamma_1 = 0.83$ (from Table 4). We assume a heterogeneous demand distortion $\gamma_i = \text{logistic}(\mu_c + \sigma_c \eta_i)$, $\eta_i \sim N(0, 1)$, with dispersion $\sigma_c = \log(1 + \exp(\Omega_4 + \Omega_5(1 - \gamma_c)))$, such that Ω_4 governs baseline dispersion and Ω_5 controls how dispersion varies with the mean wedge, allowing for heterogeneity in credit impacts to affect the dispersion of WTP.²⁰ The location parameter μ_c is pinned down by $\mathbb{E}[\gamma_i|c] = \gamma_c$. Individual WTP is then given by $wtp_i = (1 - \gamma_i)b_i$.

The distributions of wtp_i and ϕ_i can be used to calculate demand elasticity $\mathcal{E}_D$ and the average externality of adopters who are marginal to the subsidy ($\bar{\phi} = \mathbb{E}[\phi_i|p - s_h \leq wtp_i < p - s_l]$, where $p - s_h = \text{US\$20}$ and $p - s_l = \text{US\$73}$ as in the experiment).

7.2 Model estimation

We target seven empirical moments: extensive margin elasticity $\mathcal{E}_{D,c}$, average marginal externality $\bar{\phi}_c$, and cost per marginal adoption—all of which vary by credit groups $c \in \{0, 1\}$ —as well as a common $\nu = \text{corr}(b_i, \phi_i)$. For each moment n we define the simulated moment error $e_n(\mathbf{\Omega})$ as a normalized function of the empirical moment $m_n(x)$ and the simulated moment $\hat{m}_n(\tilde{x})$:

$$e_n(\mathbf{\Omega}) = \frac{\hat{m}_n(\tilde{x}) - m_n(x)}{m_n(x)}$$

We estimate the model parameters $\mathbf{\Omega} = \{\Omega_1, \Omega_2, \Omega_3, \Omega_4, \Omega_5\}$ using two-step generalized method of moments (GMM) with simulated moments, defining the estimator:

$$\hat{\mathbf{\Omega}}_{GMM} = \mathbf{\Omega} : \min_{\mathbf{\Omega}} e(\mathbf{\Omega})^T \mathbf{W} e(\mathbf{\Omega})$$

We simulate 100,000 agents per parameter evaluation, with a fixed seed for reproducibility. The first-step weighting matrix is $\mathbf{W}_1 = \mathbf{I}$; the optimal second-step weight is $\mathbf{W}_2 = \hat{\mathbf{\Gamma}}^{-1}$, where $\hat{\mathbf{\Gamma}}$ is the sample covariance of the moment-error vector across 30 bootstrap re-simulations at the first-step estimate. The reported estimate corresponds to the lowest objective value across multiple starting values and satisfies the optimizer’s convergence criterion.

The estimated coefficients are $\hat{\Omega}_1 = 1.9$, $\hat{\Omega}_2 = 1$, $\hat{\Omega}_3 = 0.6$, $\hat{\Omega}_4 = -2.7$, and $\hat{\Omega}_5 = 30$. Panel a of Table 7 compares the moments simulated with the estimated parameters and the empirical

²⁰The logit-normal distribution across different $\mathbb{E}[\gamma]$ reasonably approximates the LASSO-generated empirical distributions (Figure B4). Even though it also operates on the interval $[0, 1]$, we do not apply the Beta distribution because its behavior at the boundaries of $[0, 1]$ is inconsistent with what we observe in our data.

Table 7: Simulated estimates from observed and counterfactual parameters

(a) Estimation

Moment	RCT: Pay up front ($\gamma = 0.91$)			RCT: Pay with loan ($\gamma = 0.83$)		
	Empirical	Simulated	Deviation	Empirical	Simulated	Deviation
Correlation (ν)	0.8	0.8	-1%	0.8	0.8	-1%
Marginal externality (ϕ)	5.7	5.8	1%	4.8	4.9	1%
Extensive margin elasticity ($\mathcal{E}_D$)	-2.3	-2.4	-1%	-0.9	-0.7	17%
USD per marginal adoption	67.5	66.8	-1%	95.5	99.6	4%

(b) Counterfactual simulations: Varying the distortion γ

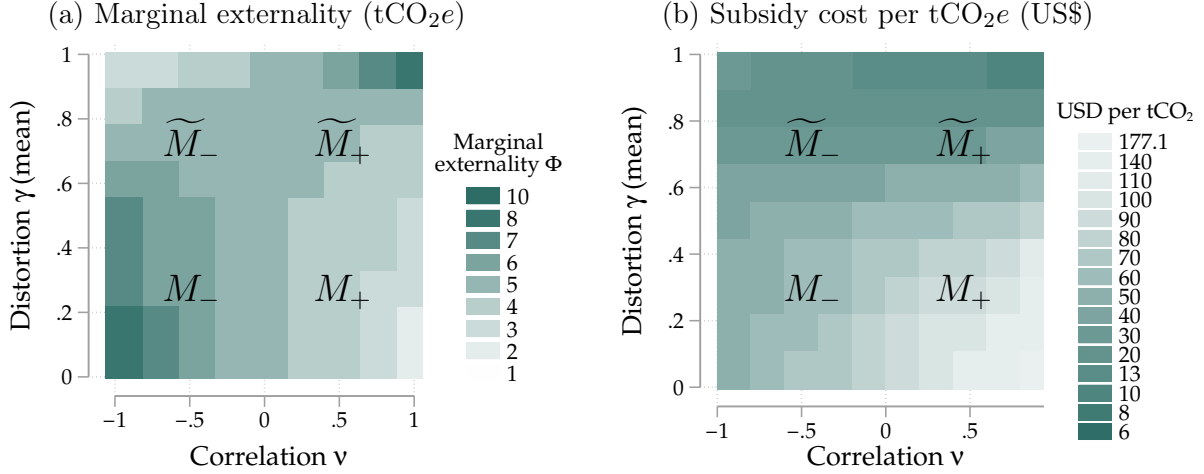
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Assumed demand distortion (γ)	0.95	0.91	0.83	0.75	0.5	0.25	0
Marginal externality (ϕ)	7.8	5.8	4.9	4.5	3.7	3.2	2.8
Extensive margin elasticity ($\mathcal{E}_D$)	-8.4	-2.4	-0.7	-0.4	-0.2	-0.1	-0.1
USD per marginal adoption	64	67	100	144	273	369	449
<i>Subsidy cost per tCO₂e (in USD)</i>							
Observed		12	20				
Simulated	8.2	12	20	32	73	114	160

Notes: Panel a compares empirical moments and the simulated moments under the estimated model parameters. There are seven moments: a single correlation moment ν , and one per treatment group for each of the other three moments. Panel b uses the model parameters to estimate key moments, as well as the aggregate subsidy cost per tCO₂e, for counterfactual scenarios with higher or lower demand distortions, holding constant the correlation between private benefits and social benefits to match the empirical moment and assuming $r = 0$. [Table B6](#) reports the same estimates with standard errors and confidence intervals. [Table B7](#) also varies the correlation parameter ν . [Table B8](#) and [Table B9](#) show that these results are not sensitive to the specific distributional assumption of b_i and γ_i .

moments calculated using the data. The subsidy costs per tCO₂e implied by the targeted moments, US\$12 and US\$20, closely match the empirically estimated values of US\$12 and US\$20, respectively. With these estimates in hand, we run counterfactuals by drawing distributions of private benefits ($b_i/\hat{b}$) $\sim$ Weibull($\hat{\Omega}_1, \hat{\Omega}_2$), social benefits $\varepsilon_i \sim$ Weibull($\hat{\Omega}_1, \hat{\Omega}_2$) and distortions γ_i from the logit-normal distribution with dispersion $\sigma = \log(1 + \exp(\hat{\Omega}_4 + \hat{\Omega}_5(1 - \gamma)))$ and location pinned down by $\mathbb{E}[\gamma_i] = \gamma$.

Counterfactual I: Demand distortion Panel b of [Table 7](#) shows the results implied by the model for different assumptions of γ . Demand distortions that dampen demand increase the positive externality generated by the marginal adopter and demand elasticity, lowering the subsidy cost per tCO₂e. In the neoclassical case where private benefits equal WTP—corresponding to no demand distortion—the subsidy cost per tCO₂e increases to US\$160. Similar patterns persist when estimating the observable welfare gain, Marginal Value of Public Funds (MVPF), and resource cost per ton ([Table B5](#)).

Figure 8: Counterfactual estimates



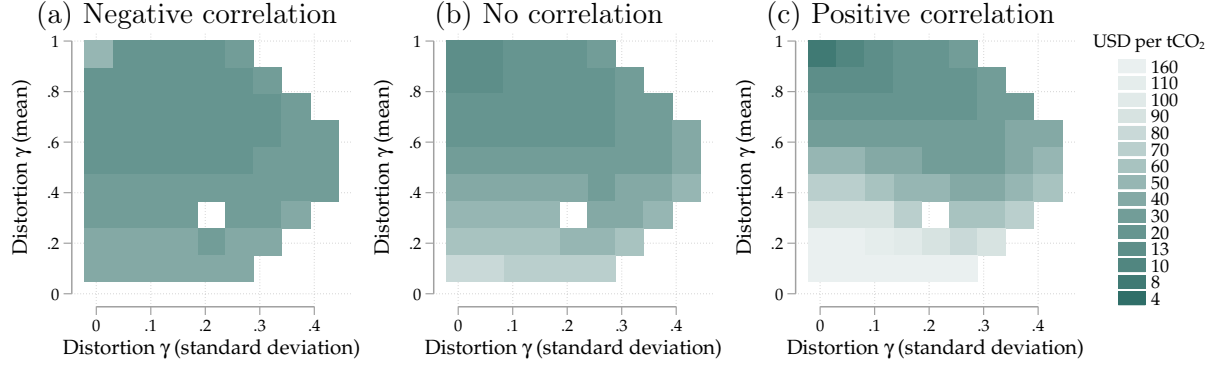
Notes: Both figures show how demand distortions as well as the correlation between private and social benefits affect subsidy efficacy. Panel a shows the marginal externality. The labels $\widetilde{M}_+$, M_+ , $\widetilde{M}_-$, and M_- correspond to the labels in panels a and b of Figure 1, which presents the theoretical underpinnings for channels 1 and 2. Panel b shows the combined effect of the marginal externality and the elasticity on the subsidy cost per tCO₂e. The demand distortion decreases the subsidy cost per tCO₂e everywhere in this context, regardless of correlation. Table B7 presents a subset of these estimates in the format of Table 7.

We estimate two alternative specifications of the model to test the robustness of these results to alternate distributions of b_i and γ_i . The first assumes linear distributions, with benefits following a downward-sloping distribution over $[0, \Omega_1]$ and the distortion distributed over $[0, 1]$ with a triangular distribution parameterized by its mode, which yields three parameters. The second uses the same distribution of benefits as the main specification but estimates a logit- t distribution instead of a logit-normal distribution. Both simulations show similar patterns across all measures of subsidy efficiency (Table B8 and Table B9).

Counterfactual II: The correlation between private and social benefits We extend the counterfactuals to include alternative correlations between private and social benefits. Figure 8 shows the results when varying not only the demand distortion γ but also the correlation ν . Positive values of ν are generated by varying the mixing weight Ω_3 ; negative values are generated by mixing in a draw from the same Weibull distribution that is rank-anti-correlated with $b_i/\hat{b}$. The horizontal axis reports the realized correlation. To help build intuition, we use labels $\widetilde{M}_+$, M_+ , $\widetilde{M}_-$, and M_- to demarcate the approximate externalities (also labeled in panels a and b of Figure 1).

When WTP reflects private benefits and there is no distortion ($\gamma = 0$, such that $wtp_i = b_i$), a positive correlation induces adverse selection, decreasing the marginal positive externality ($M_+ < M_-$). On the other hand, under a demand distortion that dampens WTP ($\gamma > 0$), a positive correlation means compliers generate larger positive externalities, inducing beneficial selection ($\widetilde{M}_+ > \widetilde{M}_-$). The effect of a demand distortion thus hinges on the correlation between private and social benefits: when this correlation is negative, a demand distortion decreases the average externality of compliers ($\widetilde{M}_- < M_-$). It is only when it is positive that the distortion increases the average externality of

Figure 9: Counterfactual estimates with heterogeneous demand distortions



Notes: This graph visualizes how different demand distortions and correlations affect the subsidy cost per tCO₂e when demand distortions are allowed to be heterogeneous across agents, independently of private and social benefits. Since we constrain $\gamma \in [0, 1]$, not all combinations of mean and standard deviation are feasible. The realized correlations in panels a and c are -0.94 and 0.94 , the most negative value attainable given the Weibull marginal distributions and its positive counterpart.

compliers ($\widetilde{M}_+ > M_+$).

Panel b shows how both channels together affect the subsidy cost per tCO₂e. While the marginal externality channel depends on whether the correlation is positive or negative, the demand distortion increases the extensive margin elasticity, regardless of the correlation between private and social benefits. In this context, this latter effect outweighs the opposing effect on the average externality of compliers among negative correlation coefficients: even though $\widetilde{M}_-$ has a lower marginal externality than M_- (panel a), it still has a lower subsidy cost per tCO₂e (panel b). As a result, the demand distortion unambiguously decreases the subsidy cost per tCO₂e in this context, regardless of the correlation coefficient ($\widetilde{M}_- < M_-$ and also $\widetilde{M}_+ < M_+$).

Counterfactual III: Heterogeneous demand distortions across agents Finally, we explore how heterogeneity in the demand distortion affects subsidy efficacy. For simplicity we assume the γ_i are uncorrelated with any of the other parameters of interest (private benefits and social benefits). We draw $\gamma_i = \text{logistic}(\mu + \zeta \eta_i)$, $\eta_i \sim N(0, 1)$, varying μ and ζ to vary means and standard deviations of γ_i while ensuring $\gamma_i \in [0, 1]$.

Figure 9 shows the results. When the standard deviation approaches 0, the graphs follow the patterns in panel b of Figure 8: the mean distortion matters more when private and social benefits are positively correlated. All three panels show that this pattern weakens as the standard deviation increases.

8 Conclusion

Demand distortions such as financial market frictions and behavioral biases are pervasive across many low- and middle-income countries, often operating by dampening willingness-to-pay (WTP) for profitable technologies. This paper shows that this has important implications for the efficiency

of Pigouvian price instruments. In theory, subsidies for green technologies generate a larger gain in observable welfare per subsidy dollar as well as a larger positive externality when agents face demand distortions that dampen demand. This operates through two channels: increasing the private and social benefits of the marginal adopter, and increasing demand elasticity.

We test this theory in a field experiment with more than 2,000 households in Nakuru County, Kenya, where we randomize marginal-cost subsidies, fixed-cost subsidies, and demand distortions for the purchase of an energy efficient technology: induction stoves. We first use the random variation in prices to estimate the causal impacts of stove adoption. We find that the stove abates 5.5 tCO_{2e} and reduces (discounted) household energy expenditures by US\$209 over the 2.2 years of expected usage of the stove. WTP among the control group is US\$20: access to a loan increases WTP to US\$35. Even factoring in continued use of charcoal stoves and imperfect additionality of the subsidies, green subsidies in this context offer a very low-cost abatement opportunity, at only US\$12 per ton of CO_{2e} abated. This compares favorably to many other abatement technologies available today. Factoring in the private benefit to adopters, the externality benefit, and the stove manufacturing cost, each dollar of subsidy generates between US\$10–US\$12 in observable welfare benefit.

As predicted by the theory, demand distortions in this context contribute to these favorable numbers. The randomly assigned access to credit, which lowers the demand distortions, lowers the observable welfare benefit per subsidy dollar to US\$7–US\$8 and raises the subsidy cost per tCO_{2e} to US\$20. This operates through the two channels identified in the model: demand distortions increase the marginal adopter’s positive externality by 19% and increase demand elasticity from -0.9 to -2.3 .

The loan treatment increases WTP in this context but still does not close the adoption gap. To estimate the subsidy cost per tCO_{2e} in the absence of any demand distortions, we estimate the model using two-step generalized method of moments (GMM), using the random variation from the experiment to identify key parameters. Counterfactuals suggest that if WTP equaled the discounted stream of fuel savings the subsidy cost per tCO_{2e} would have increased more than tenfold, to US\$160.

The mechanisms identified in this paper suggest that many of the most effective abatement opportunities may be found in contexts with larger demand distortions. As an example, the U.S. has relatively sophisticated financial markets for consumer durables, with the vast majority of new cars sold with financing. Our paper suggests that this could contribute to the high subsidy cost per tCO_{2e} of electric vehicle subsidies, at more than US\$1,200.

Approximately 1.4 million Kenyan households use charcoal on a daily basis. Optimistically, a price subsidy could incentivize adoption by up to 37% of households.²¹ Peak-hour usage at the 90th percentile in our sample is 88 Wh per customer, so scaling our experimental results would require an additional 45 MW in grid capacity, against Kenya’s current peak demand of 2,177 MW and

²¹This assumes universal electricity access among charcoal users, which is not completely unreasonable. Charcoal is used primarily in urban and peri-urban areas and urban electricity access was 91% in 2024 (KNBS, 2024).

installed capacity of 3,243 MW (Kenya Power, [2024](#)). Kenya’s electric utility appears capable of meeting the demand increase associated with widespread electric stove adoption.

Approximately 54 million households across the world today use charcoal as their primary cooking fuel, including approximately 31 million households that live in Africa (Berkouwer and Dean ([2026c](#))). At annual abatement of 2.6 tCO_{2e} and with 37% adoption at subsidized prices, a widespread cooking transition in Africa would abate approximately 30 million tons of CO_{2e} per year. In addition, more than 300 million households across the world still cook primarily with firewood, and the IEA ([2025](#)) estimates that the number of households cooking with wood or charcoal in Africa in 2050 will, if anything, be higher than today.

More than US\$1 trillion is invested towards climate change mitigation each year, much of which is targeted at rapid electrification combined with grid decarbonization. Maximizing the impact of these expenditures requires allocating this funding towards abatement opportunities at the lowest portion of the abatement cost curve. Given the results of this study, many of these opportunities may be found in low- and middle-income countries, where demand distortions are pervasive. Additional research is needed to identify specific mitigation opportunities, and to understand how the heterogeneity of demand distortions across countries affects global cooking transition pathways.

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